



Investigation of Sedimentary Indicators of Eutrophication and Trace Metal Pollution in the Southern Marmara Sea (Türkiye)

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Abstract This study evaluates the spatial and historical patterns of primary productivity proxies and trace elements in core and surface sediments from the southern Marmara Sea following the 2020 mucilage event, with emphasis on the Erdek and Bandırma Gulfs. Stratigraphic records reveal a sustained rise in eutrophication indicators and metal enrichment over the last ~200 years, coinciding with accelerated industrialization and urban expansion in the region. Surface sediments from the southern Erdek Gulf display pronounced accumulation of nitrogen, phosphorus, biogenic silica, organic carbon, and trace metals (As, Cd, Cr, Ni, Zn, Hg), primarily driven by

terrestrial inputs associated with the Gönen River catchment. In the Bandırma Gulf, elevated contamination levels are attributed to maritime traffic, port operations, and industrial discharge, with the western shoreline representing the most impacted sector while northern coastal areas remain comparatively less contaminated. Anthropogenic enrichment of Cu, Cd, Zn, Ni, As, Cr, and Hg underscores the dual vulnerability of these systems to eutrophication and chemical pollution. Furthermore, this work provides the first characterization of dinoflagellate cyst assemblages during a mucilage episode in the area, identifying 15 taxa including harmful species (*Alexandrium* sp., *Operculodinium centrocarpum*, *Lingulodinium machaerophorum*). Their presence indicates elevated ecological risk and potential for future harmful algal bloom development, particularly in the Bandırma Gulf.

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1 Introduction

River mouths and coastal ecosystems, in general, are recognized as highly productive and dynamic areas, providing essential support for a significant portion of the global population as well as evolving drainage basins (Paerl, 2006). The progression of human activities in coastal regions and basins has resulted in increased human-induced pressures on the ecosystem, stemming from inputs such as land-based nutrients, trace metals, wastewater, herbicides, and pesticides (Y. Wang et al., 2017). Coastal regions hold particular significance as they serve as interfaces between marine and freshwater environments. The intricate biological, chemical, and physical exchanges between seawater and freshwater play a crucial role in shaping the characteristics of suspended particles containing various elements (Zhai et al., 2021). Owing to the intricate interplay of marine and fluvial processes, river mouths function as geochemical repositories for metals that are bound to sediments (Wang et al., 2022). Sediments are widely acknowledged as reservoirs for metals, offering historical perspectives on metal inputs and presenting numerous benefits for climate and environmental evaluations (Xie et al., 2021). Furthermore, sedimentary records offer crucial insights into pollution trends that have originated in the past and strategies for their mitigation. Thus, sediments are favored as a monitoring tool due to their relatively stable nature over short temporal scales compared to water (Atkinson et al., 2007).

Certain elements known as Potentially Toxic Elements (PTEs) possess the capability to induce toxicity within ecosystems and can infiltrate the marine food web via water, food sources, and sediments, ultimately posing risks to human populations (Shahabi-Ghahfarokhi et al., 2021; Wang et al., 2022). In addition to their detrimental effects on human health, these elements also jeopardize the diversity and abundance of various organisms (Wang et al., 2022). Consequently, numerous elements, particularly Mercury (Hg), Cadmium (Cd), Lead (Pb), and Arsenic (As), are subject to extensive examination and surveillance to mitigate their adverse consequences (Fural et al., 2020; Kükreer et al., 2020; Özkan et al., 2022;

Shahabi-Ghahfarokhi et al., 2021; Ustaoglu, 2020; Ustaoglu & Islam, 2020).

Silicon, nitrogen, and phosphorus are vital organic components necessary for the growth of phytoplankton, energy transfer in the food chain, and the operation of coastal marine ecosystems (Dan et al., 2021; Kükreer et al., 2021). Excessive nitrogen and phosphorus inputs induce unforeseen alterations in primary production, stimulate algal blossoms, decrease water clarity and dissolved oxygen levels, harm the structure of the food chain, and can instigate toxicity episodes (Zhang et al., 2019). Harmful algal blooms (HABs), a phenomenon stemming from environmental deterioration and posing a risk to both marine ecosystems and public health, are also a point of concern. Among 4000 recognized marine microalgae, 200 are detrimental, and 80 species are categorized as poisonous (Balkis et al., 2016). Certain microalgae species can exist in cyst form during a segment of their life cycle. The transition from vegetative cells to cysts occurs under diverse environmental conditions, and cysts can persist in sediments for many years. Under favorable conditions, they can revert to vegetative form and start to propagate, triggering HABs (Balkis et al., 2016). Thus, investigating cysts in sediments is critical as a species not currently found in the water column could pose a future hazard to the ecosystem. Understanding the chemical and biological makeup of sediments in coastal areas, influenced by ongoing population growth, industrial expansion, sewage and liquid waste, pesticides, fertilizers, ship traffic, and discharges from marine and floating structures, is crucial for forecasting the destiny of these released pollutants (Badawy et al., 2021; Yang et al., 2021).

Bandırma is a prominent Turkish port city. The port can host 330 container ships and various cargo vessels annually, operating near full capacity.¹ Heavy industrial facilities, producing goods such as fertilizer and automotive products, are located south of Bandırma Gulf. Additionally, domestic waste from Bandırma is discharged into the Marmara Sea. Key human activities beyond the port include airport operations and agriculture. In the northern Bandırma Gulf region, tourism is the main human activity. This area lacks industrial or agricultural influences. Erdek district, with a population of 31,902, is a major tourism

¹ <http://www.portofbandirma.com.tr/kurumsal/liman-hakkinda>

center in the northern Erdek Gulf (TÜİK, 2022).² To the north of Erdek district, agricultural areas primarily consist of olive groves. Agriculture is the dominant human activity in the southern Erdek Gulf. The Gönen River basin, draining into the Marmara Sea, irrigates Turkey's major rice cultivation areas. The Gönen River also carries domestic and industrial waste from the 74,871 residents of Gönen district (TÜİK, 2022) into Erdek Gulf. Intense anthropogenic impacts in these gulfs led to a mucilage event in the summer of 2020, caused by excessive algal growth.

Since coastal ecosystems are fragile environments, it is crucial to approach pollution studies from multiple perspectives in order to capture the various dimensions of the problem. In the study area, a lack of multidisciplinary research has been identified—specifically studies that simultaneously examine the temporal and spatial distributions of organic and inorganic pollutants along with the phytoplankton cysts, which represent the first group of organisms directly affected by these processes. This study addresses that gap by presenting a pollution profile across both spatial and temporal scales, and by compiling an inventory of cysts present in the sediment that, under favorable future conditions, may germinate and proliferate in the water column.

The aims of the study conducted during this period are as follows: (i) to determine the temporal and spatial distribution PTEs and eutrophication markers such as total nitrogen (TN), total phosphorus (TP), chlorophyll degradation products (CDP), total organic carbon (TOC) and biogenic silica (BSi) accumulated in sediments; (ii) to identify ecological risks stemming from these variables; (iii) to assess pollution trends and the processes controlling them; and (iv) to identify dinoflagellate cysts in sediments to make predictions for potential HABs in the future.

2 Material and Methods

2.1 Sediment Sampling and Analysis

A total of 10 surface sediment samples were gathered using a Van Veen Grab, comprising 5 samples

from Bandırma Gulf and 5 samples from Erdek Gulf in June 2020, in the period of mucilage. One sample was taken from each station, homogenized thoroughly to ensure representativeness, and subsequently processed for analytical procedures. Simultaneously, a 20 cm core was extracted from the mouth of the Gönen River (40 ° 20' 44.27 N; 27 ° 37' 59.88E) in the southern region of Erdek Gulf to assess the temporal changes in element concentrations and background values, employing a gravity core sampler (Fig. 1).

Subsequently, the core was segmented into 2 cm slices – total 10 samples- for further analysis. The acetone extraction method (Lorenzen, 1971) was applied to sediment samples to conduct CDP analysis. In this method, wet sediment samples were extracted in 90% acetone and centrifuged to obtain pigment extracts. The resulting supernatant was measured spectrophotometrically to quantify CDP. Following this, the sediment samples underwent drying in an oven at 60 °C, grinding in a porcelain mortar, and sieving through an 11 mesh sieve. The remaining analyses were performed on oven-dried samples. TOC content of sediment samples was determined using the Walkley–Black dichromate oxidation method. In this method, organic matter is oxidized by potassium dichromate in an acidic medium, and the remaining dichromate is titrated with ferrous ammonium sulfate. The amount of oxidized organic carbon is then calculated based on the volume of titrant used. (Gaudette et al., 1974). Carbonate content was determined using a Scheibler Calcimeter following the method of Schlichting and Blume (1966). The technique is based on the volumetric measurement of CO₂ released by the reaction of carbonate with hydrochloric acid. BSi content was determined using the alkaline extraction method described by DeMaster (1981) In this procedure, samples are treated with a sodium carbonate solution at elevated temperature to dissolve the BSi. The concentration of dissolved silica is then measured spectrophotometrically. The elemental analyses were conducted via ICP-MS at Bureau Veritas Analytical Laboratory (Canada), with the inclusion of reference materials, duplicate measurements, and blank samples to verify accuracy. Prior to analysis, samples were digested with a mixture of strong acids under controlled conditions. The recovery rates for the elements were determined to fall within the range

² <https://data.tuik.gov.tr/Bulten/Index?p=Adrese-Dayali-Nufus-Kayit-Sistemi-Sonuclari-2022-49685>

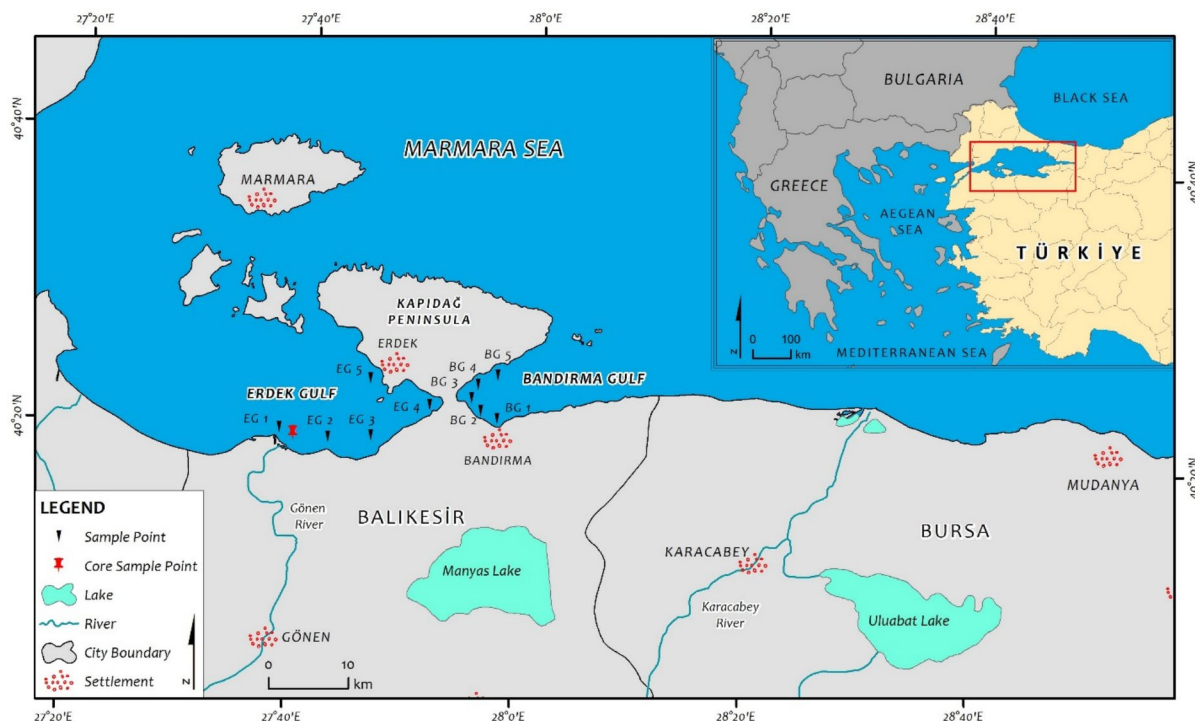


Fig. 1 Geographic map of the study area showing sample sites

of 93.7% to 101.1%. The analysis of TN content was completed at the Central Laboratory Application and Research Center of Kastamonu University using a CHNS elemental analyzer. In this method, samples are combusted at high temperatures in an oxygen-rich environment, and the resulting gases are separated and quantified. All maps produced in the study were prepared with the Kriging interpolation method using Arc—GIS Pro software. The data from 10 sampling points were directly entered and were classified and mapped using the Graduated Symbols tool in the Symbology interface of ArcGIS Pro software.

2.1.1 Calculation of Ecological and Toxicological Risk Indices

The utilization of the enrichment factor (EF) and the geoaccumulation index (Igeo) served the purpose of distinguishing between the natural and

anthropogenic origins of elements. EF was computed through the utilization of a specific mathematical expression denoted as Formula 1.

$$EF = \frac{\left(\frac{C_i}{C_{ref}}\right)_{sample}}{\left(\frac{B_i}{B_{ref}}\right)_{background}} \quad (1)$$

Within this formula, C_i denotes the concentration of the element, C_{ref} signifies the concentration of the element utilized for geochemical standardization, B_i represents the background value of the element, and B_{ref} indicates the background value of the element used for geochemical standardization (Brady et al., 2015; Zhang et al., 2007).

Igeo was derived through the application of Formula 2. Within the framework of this formula, C_m stands for the concentration of the element, B_m denotes the background value of the PTE, and a constant coefficient of 1.5 is integrated (Muller, 1969).

$$I_{\text{geo}} = \frac{C_m}{(B_m * 1.5)} \quad (2)$$

The computation of the toxic risk index (TRI) was undertaken to assess the deleterious impacts of elements (G. Zhang et al., 2016). Formula 3 was employed to establish the toxic risk coefficient (TRI_i) for each element.

$$TRI_i = \sqrt{\frac{((C_i/TEL)^2 + (C_i/PEL)^2)}{2}} \quad (3)$$

In this particular formula, C_i represents the element concentration, TEL (threshold effect level) indicates the threshold effect level, and PEL (probable effect level) signifies the probable effect level. The determination of TEL and PEL was conducted based on the thresholds of toxic effects detected in diverse experimental organisms and sediments (MacDonald et al., 1997, 2000).

The computation of the toxic risk index was accomplished through the utilization of Formula 4. Within this formula, TRI_i denotes the toxic risk coefficient of an individual element, n indicates the quantity of elements involved in the computation, and TRI reflects the toxic risk level (G. Zhang et al., 2016).

$$TRI = \sum_{i=1}^n TRI_i \quad (4)$$

In order to evaluate the ecological risk levels associated with elements, the modified ecological risk Index (mRI) metric was employed. The mRI serves as a representation of the ecological risk level for each element. By summing the mRI values, integrated modified ecological risk Index (mPER) is determined. Ecological risk and potential ecological risk assessments have extensively relied on contamination factor (CF) data for several years (Hakanson, 1980). Nevertheless, due to the absence of geochemical normalization in CF calculations, errors stemming from grain size variations remain unaddressed. Consequently, the development of mRI and mPER based on EF data and Formula 5 was introduced (Brady et al., 2015). In this formula, Tr_i represents the toxic risk coefficient of elements. The toxic risk coefficients assigned to the elements are as follows: Hg=40, Cd=30, As=10, Cu=Pb=Ni=5, Cr=2, Zn=1, (Hakanson, 1980).

$$mRI = EF \times Tr_i \quad (5)$$

The calculation of mPER, which aggregates the mRI values of the PTEs, was carried out utilizing Formula 6.

$$mPER = \sum_{i=1}^n mRI \quad (6)$$

A detailed outline of the evaluation scale for ecological risk indices is delineated in Supplementary data 1.

Table 1 Results of AMS radiocarbon analyses

Depth (mm)	Conventional Age (yr BP)	Calendar Age Range (cal yr BP)	Materials
90	550 ± 23	203–147	Bulk
190	1120 ± 27	710–636	Bulk
Depth (mm)	Mean Calendar Date (cal yrs BP)	20 mm Age Range (Calendar Date)	Sedimentation Rate (mm/yr)
0–20	2000	2020–1980	0,50
21–40	1955	1977–1930	0,40
41–60	1902	1928–1872	0,34
61–80	1839	1869–1805	0,30
81–100	1767	1801–1728	0,26
101–120	1686	1724–1643	0,23
121–140	1596	1638–1548	0,21
141–160	1497	1543–1444	0,19
161–180	1388	1439–1331	0,18
181–195	1287	1326–1241	0,16

2.1.2 Dating and Age-Depth Model

Accelerator Mass Spectrometry (AMS) ^{14}C analyses were performed on two marine sediment samples at the TÜBİTAK-MAM Radiocarbon Laboratory, as presented in Table 1. The sediment core was retrieved using a gravity corer, a method that effectively minimizes disturbance of the uppermost sediments and thus preserves the stratigraphic integrity of the core top. Consequently, the top of the core was assigned an age corresponding to the time of core recovery, which was incorporated as the most recent chronological constraint in the age-depth model. This practice is widely employed for short gravity cores obtained from subaqueous depositional settings (e.g., Biltekin et al., 2023, 2025; Çağatay et al., 2012; Erginal et al., 2019; Makaroğlu et al., 2022; Meydan et al., 2024, and references therein). In addition to the core-top constraint, two radiocarbon determinations from the 8–10 cm and 18–20 cm depth intervals were utilized for the development of the age-depth model. Radiocarbon ages were calibrated using the Marine20 calibration curve (Reimer et al., 2020) (Table 1). All results are corrected with the marine reservoir ages ($R(t)$) 390 ± 85 (ΔR) years (Siani et al., 2000) (Supplementary data 2).

Age-depth modelling was conducted using the non-Bayesian *Clam* script implemented in RStudio, which estimates sediment accumulation chronology

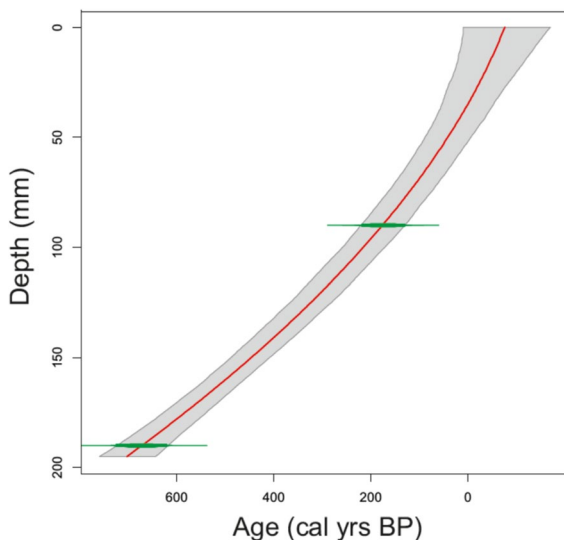


Fig. 2 Age-depth model

within a 95% Gaussian confidence envelope, following the approach proposed by Blaauw, (2010) (Fig. 2). Accumulation rates/sedimentation rates are calculated by the age-depth model using the cubic spline iteration of mm scale samples (Supplementary data 2). In order to accurately correlate the physical sampling and their corresponding ages, we also provided mean age range of each physical 2 cm long samples (Table 1).

2.1.3 Sediment Sampling and Preparation for Dinoflagellate Cyst

Surface sediments were obtained once using a Van Veen Grab from 3 stations in Erdek Gulf and 2 stations in Bandırma Gulf. The methodology proposed by Matsuoka and Fukuyo, (2000) was employed for the treatment of samples intended for dinoflagellate cyst analysis. Collection of the top 2 cm of sediment was conducted, followed by storage at 4 °C in the absence of light until further preparation.

In the context of dinoflagellate cyst analysis, the sediment samples were assessed in their wet state and subjected to treatment with hydrochloric acid (10 ml ca. 10% HCl) and hydrofluoric acid (10 ml ca. 47% HF) at room temperature to dissolve calcium carbonate and silicate, respectively. Post-treatment, the residue underwent multiple rinses with distilled water to eliminate residual acids until reaching a pH of approximately 7. Subsequently, the processed samples underwent sonication for 30 s using a KUDOS SK1200H Ultrasonic Cleaner. The resulting residue was then sieved through stainless steel meshes with pore sizes of 125 μm and 20 μm , and suspended in 10 ml of distilled water within a vial. Identification procedures were carried out on 1 ml of the prepared sample stored in a vial, utilizing a Sedgewick-Rafter counting chamber under an inverted microscope (Olympus IX71-DIC attachment) equipped with a digital camera (Olympus DP25) at 400 $\times$ magnification.

The terminology utilized for cysts was primarily paleontological, unless specific cysts were solely known by their biological name. Cysts attributed to *Alexandrium* spp. were designated based on type or complex due to the morphological similarities in cyst types produced by different species of *Alexandrium*.

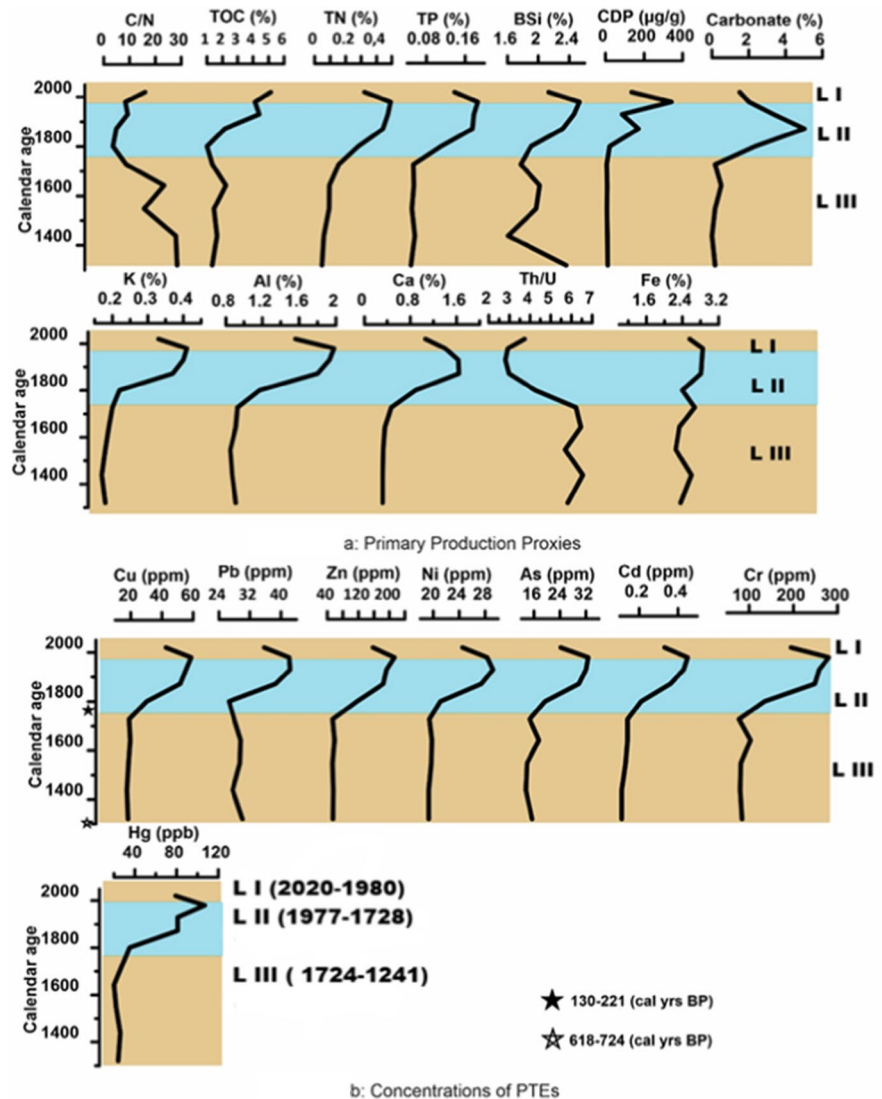
3 Results and Discussions

3.1 Temporal Variations and Accumulation Processes of Eutrophication Markers

The distribution of primary production proxies derived from the Erdek Gulf core emphasizes three distinct periods (Fig. 3a). The time span from 1980 to 2020 (L I) signifies processes predominantly influenced by terrestrial carbon. Notable trends in this time interval reveal a similar pattern in carbonate and Ca levels, as well as a typical decrease in TN, TP, BSi, CDP, Fe, K and Al contents. These changes suggest a decrease in nutrient inputs. Although all

the palaeoclimatic proxies clearly reflect arid conditions, the relatively high sedimentation rate (S 2) at this level, as indicated by the AMS14C data, is likely related to increased sediment supply during short-lived but intense rainfall events that commonly occur under such dry climatic conditions. Terrestrial mineral elements (K, Al, Fe) are transported by high-energy, long-duration precipitation events, while terrestrial organic matter—such as leaves, woody debris, and soluble humic substances, which are lighter materials—can be transported by short-duration rainfall events. Especially ephemeral streams accumulate terrestrial coarse particulate organic matter while dry, flushing it into downstream environments during

Fig. 3 Temporal variation of primary production proxies and PTEs concentrations



rainfall events (Fritz et al., 2019). In this time interval, the C/N ratio reaches 16.02, indicative of terrestrial carbon contributions. Scant amounts of TN and TP, primarily originating from agricultural regions, imply arid conditions during this epoch and weak nutrient transfer from farmlands to the ocean, resulting in diminished plant productivity.

The Gönen River Basin covers an area of 913 km². Of this total area, 369 km² is occupied by agricultural and residential land, while the remaining portion consists of natural biomes. Within the basin, there are 268 km² of agricultural land and 101 km² of residential areas. Thus, approximately 30% of the entire basin is utilized for agricultural purposes. Digital data obtained from Landsat satellite imagery indicate that agriculture is the predominant anthropogenic activity in the basin (Özşahin, 2011). The interval between 1728–1977 (L II) characterizes the subsequent phase dominated by fluvial transport, associated nutrient inflows, and marine production mechanisms. Estuarine zones are marked by the mingling of seawater and freshwater, substantial nutrient fluxes, and elevated primary production (Liu et al., 2017). Notably, Fe, Al, and K concentrations rise at L II, while the C/N ratio declines to levels below 10. These indicators suggest an algal source for carbon and an increased influx of terrestrial erosion materials. This timeframe coincides with periods of heightened precipitation, as indicated by the increased concentrations of K, Al, Ca, and Fe. Concentrations of TN, TP, BSi, and CDP exhibit an upward trajectory during this time interval, reflecting amplified accumulation of TN, TP, and TOC in the sediment likely owing to algal production and the deposition of deceased algal cells.

The period spanning from 1241 to 1724 (L III) is recognized as a phase characterized by relatively subdued terrestrial erosion. Stable TP and CDP values are observed, with a marginal uptick in TN concentration. Despite the diminished terrestrial inputs during this period, the elevation in the C/N ratio might be attributed to nitrogen losses during diagenesis within the sediment (Choudhary et al., 2018). The Redfield ratio is a biochemical ratio indicating that marine organisms, on average, utilize carbon (C), nitrogen (N), and phosphorus (P) in a proportion of 106:16:1. It serves as a fundamental reference for understanding nutrient cycles and the sources of organic matter in the oceans. The N/P ratio consistently remains

below the Redfield ratio across the core, indicating prevailing nitrogen-limiting conditions in the research area (Supplementary data 3). Harrison et al., (1977) proposed that an N/Si > 1 and Si/P < 3 suggest silicon deficiency. The ratios in this study outside these thresholds throughout the core imply that silicate levels did not decrease to levels inhibiting diatom proliferation.

As per the Sediment Quality Guidelines (SQGs) threshold values established by the Canadian Ministry of Environment and Energy, sediment is considered uncontaminated when TN ≤ 0.055 and TP ≤ 0.060% while TN ≥ 0.48% and TP ≥ 0.2 indicate severe contamination (Dan et al., 2021). Consequently, TP does not indicate contamination in sediment units older than the level dated to 1728–1801 cal BP, whereas above this level, contamination levels increase, although they do not reach the threshold of significant contamination. In terms of TN, the sediment core shows contamination throughout except at the bottom, surpassing the severe contamination threshold near the surface levels (AD 1930–1977 (Supplementary data 3)). Table 2 provides the minimum, maximum, and average values for all variables. A separate study on core samples from Erdek and Bandırma Gulfs reveals that the average TOC values from Bandırma Gulf are comparable, but higher in Erdek Gulf in our study (Özen et al., 2023).

Thorium (Th) tends to accumulate in sediment in the presence of oxygen, while uranium (U) is sequestered in sediment rich in organic matter under reducing conditions (Orhan et al., 2021; Wignall & Twitchett, 1996). Therefore, a high Th/U ratio serves as an indicator of oxic conditions (Orhan et al., 2021). In this study, the decline in Th/U ratios within the sediment interval dated to AD 1728–1977 coincided with an increase in primary production, as indicated by TOC, BSi, and CDP, in the Erdek Gulf core. This decrease suggests a significant decrease in bottom water oxygen levels during the decomposition process of escalating organic matter. Along the core, an opposite trend was observed between primary production levels and the Th/U ratio, indicating a substantial influence of primary production on oxygen concentrations. Research on the evolution of hypoxia levels revealed that variations in oxygen levels over the past three centuries are linked to rising urbanization and phosphorus influxes (Jenny et al., 2016). Data from Erdek Gulf further support this notion. Our analysis

Table 2 Minimum, maximum and average values of variables

	Core sample			EG* Surface sediment			BG* Surface sediment		
	Min	Max	Average	Min	Max	Average	Min	Max	Average
TOC (%)	1.0	5.140	2.48	0.91	5.140	2.31	0.71	1.84	1.36
CDP (ug/g)	6.51	340.61	79.87	9.80	135.06	54.04	4.16	82.37	38.79
BSi(%)	1.61	2.53	2.11	2.02	3.83	2.52	1.48	2.89	2.15
TN(%)	0.047	0.492	0.246	0.118	0.414	0.254	0.040	0.211	0.106
TP(%)	0.048	0.186	0.105	0.044	0.174	0.101	0.028	0.110	0.060
Carbonate(%)	0.00	0.50	0.20	1.67	23.45	7.17	0.08	14.07	5.07
Cu (ppm)	17.51	58.75	33.16	3.27	92.9	42.65	1.41	99.9	33.004
Pb (ppm)	26.61	42.34	33.00	4.4	41.93	24.462	2.61	35.5	13.75
Zn (ppm)	53.20	213.20	113.86	17	188.3	104.32	15.6	133.6	56.52
Ni (ppm)	19.30	29.20	22.80	2.6	101.4	42.92	1.1	67.7	25.66
As (ppm)	13.40	32.70	21.28	3.2	38.4	18.48	1.4	36.4	12.78
Cd (ppm)	0.11	0.45	0.24	0.02	0.44	0.22	0.04	0.72	0.3
Cr (ppm)	77.10	280.10	154.06	5.3	269.6	133.88	1.5	73.1	26.9
Hg (ppb)	20.00	107.00	50.30	0.015	0.109	0.0516	7	381	108.4
Al (%)	0.85	1.98	1.29	0.27	2.44	1.558	0.22	1.32	0.596
Fe (%)	2.25	2.85	2.57						
Ca (%)	0.32	1.64	0.84						
K (%)	0.17	0.41	0.27						

*EG: Erdek Gulf
 *BG Bandırma
 Gulf

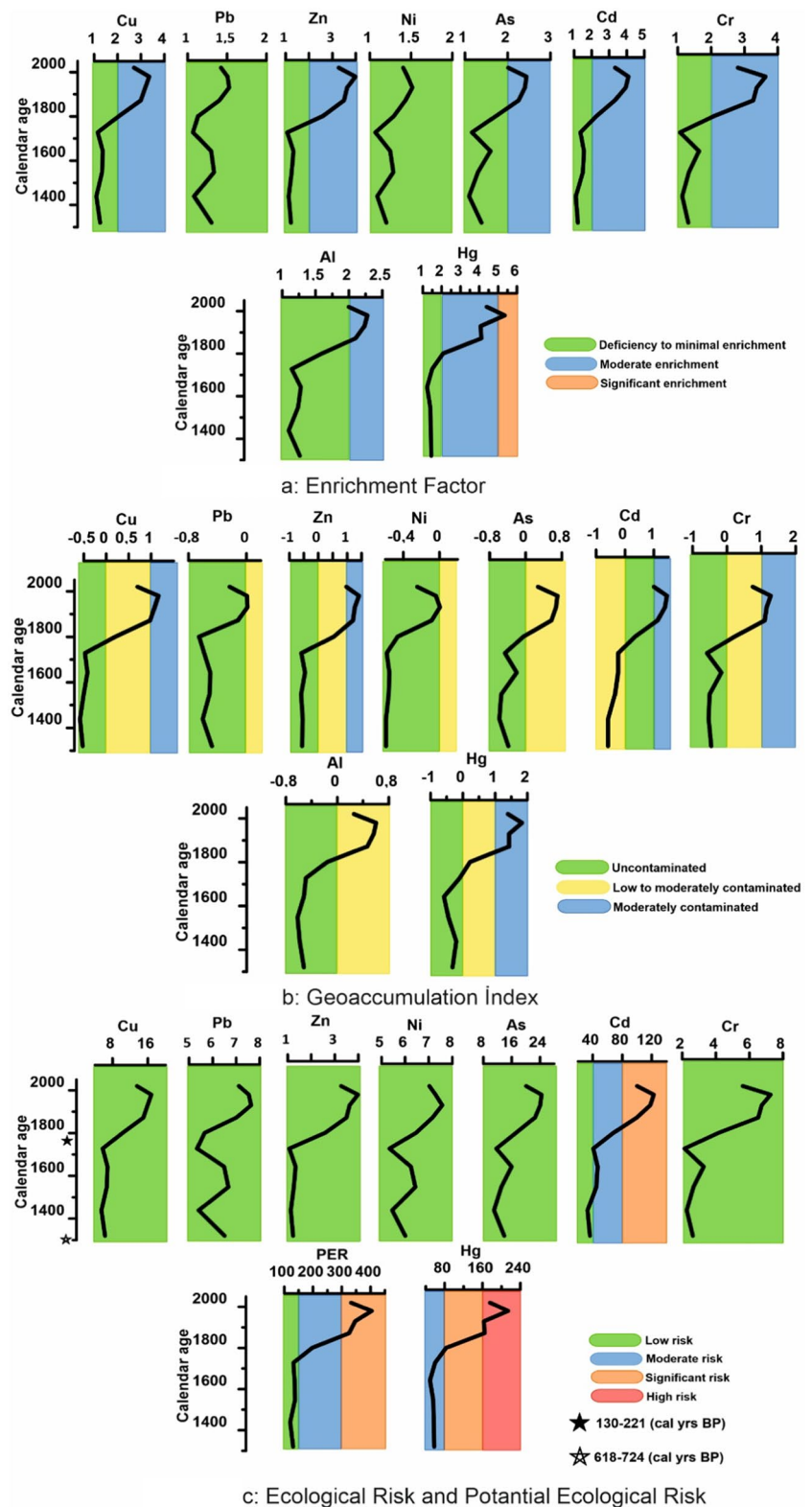
shows a negative correlation between TN, TP, CDP, BSi data, and Th/U. The Th/U enrichment observed within the sediment interval dated to AD 1980–2000 correlates positively with TOC, contrasting with the trend recorded in the lower sections of the core. This indicates that the mechanisms responsible for terrestrial carbon flux also facilitate the transport of lithogenic Th into the depositional environment.

3.2 Temporal Variations and Accumulation Processes of PTEs

An assessment was conducted on the temporal distribution of eight elements, the EF, and potential ecological/ecotoxicological risks within the core retrieved from Erdek Gulf, highlighting potential toxicity risks to the ecosystem. Analysis of scatter plots revealed a period of stable accumulation for all elements in the sediment units older than the interval dated to AD 1728–1801, succeeded by a phase of escalation starting from these levels until the dates between AD 1930–1977, possibly indicating increased anthropogenic impact. A declining

trend is currently observed in the last several decades (Fig. 3b). A comparison with a core sample from a similar sides showed higher values of Zn, As, Cr, and Cu, while Ni values were lower in our study (Kam & Önce, 2016). The identification of natural versus anthropogenic sources of PTEs often relies on two widely used tools—the EF and the Igeo. The evolution of calculated EFs across the core aligns with the raw distribution data for all elements (Fig. 4). The enrichment pattern initiated in AD 1728–1801 period persisted until the AD 1930–1977 time span, followed by a subsequent decrease. Despite Pb and Ni enrichments trending towards the surface, they did not surpass the > 2 threshold, indicating a lithogenic origin with transportation via erosion to the gulf. Considerable increment in Cu, Zn, Cd, Cr, As and Hg were observed in the same time interval. While elements enriched in the upper core section showed moderate levels overall, Hg exhibited a significant enrichment during the AD 1930–1977 period. Conversely, the older horizons of the core suggest a natural provenance for the elements. A separate study in the region highlighted anthropogenic initiation of As,

Fig. 4 Temporal variation of EF, Igeo, mRI and mPER



Hg, Pb, and Zn enrichments in the upper core section approximately 400–500 years ago (Özen et al., 2023). The same study attributed Cu, Ni, and Cr enrichments to the leather industry, indicating anthropogenic influences from the late 1800s, consistent with TN and TP data. Elements were ranked based on their mean EF values as follows: $\text{Hg} > \text{Cd} > \text{Zn} > \text{Cr} > \text{Cu} > \text{As} > \text{Pb} > \text{Ni}$ (Fig. 4).

Igeo values are in agreement with EF values (Fig. 4b) in general. The origins of Ni and Pb have been identified as natural sources with $\text{Igeo} < 0$ throughout the entire time span of the core. Prior to the dates of 19 century, all elements are considered to be at the "uncontaminated" level. Following this period, human-related element accumulation begins and extends to the present. In the past two centuries, which mark the onset of anthropogenic accumulation, As is detected at the "uncontaminated -moderate" level, whereas the majority of other elements show predominantly moderate contamination.

As indicated by the mRI values, distinct differences are observed before and after the AD 1643–1724 period, with the exceptions of Pb and Ni. There is a notable increase in risk levels from this time span. Pb and Ni demonstrate a consistent distribution throughout the time span on the core. Risk levels for all elements, except Cd and Hg, are classified as low-risk due to mRI being < 40 . Conversely, Cd and Hg pose a risk that could endanger ecosystem health. Cd has been present at hazardous levels since the 15. century while Hg remains at hazardous levels throughout the core. Cd displays moderate risk between 1500–1800, with a significant increase in risk from 19. Century to the present. Hg exhibits higher risk values, with high risk noted from AD 1805–1869 time span to present, and a gradual regression to moderate risk levels in older sediment horizons. (Fig. 4c).

The mPER values indicate a significant risk level since the 19. century within the core. While a moderate risk is identified in the sediment dated to AD 1728–1801, the older segments are characterized as low risk. (Fig. 4c).

TRI values, determined based on TEL and PEL values from ecotoxicological studies, point to risk in a substantial portion of the core. The time span AD 1728–1801 marks a shift in risk levels. Below this threshold, risks are low (TRI values range 5.96–8.81), while above it, moderate risk is evident, with the exception of the subsurface stratum where

a considerable risk is recognized (TRI values range between 11.91–16.14). (Supplementary data 3) The average TRI value (9.84) suggests a low risk in sediment samples from the core. As and Cr are the primary contributors to TRI at 32%, trailed by Ni (10%), Cu, Pb, Zn (7%), Cd (3%), and Hg (2%).

3.3 Spatial Distribution and Accumulation Processes of Eutrophication Markers

In order for regional and national policymakers to formulate and execute an efficacious management strategy, it is imperative to identify local sources of pollution (Costantini et al., 2021). To facilitate this objective, a regional evaluation of pollutant sources was conducted. Proxies for eutrophication, namely TN, TP, TOC, BSi, and CDP, were observed to be markedly concentrated along the southern periphery of Erdek Gulf. Regarding land utilization, agricultural zones are predominant along the southern shores, supplemented by inflow from the Gönen River. The levels of TOC (5.14%), BSi (3.83%), and CDP (135.06 $\mu\text{g/g}$) attain their zenith at the estuary of the Gönen River. Similarly, TN (0.41%) and TP (0.17%) also exhibit their peak concentrations along the southern coastline (Fig. 5). Unrestrained augmentations in nitrogen and phosphorus concentrations precipitate coastal eutrophication, phytoplankton blooms, and hypoxic conditions, with agricultural sectors posited as the primary contributors to these escalations (Costantini et al., 2021). The C/N ratios suggest that the Gönen River transports terrestrial carbon into the gulf. The observed < 5.5 C/N ratios at sampling sites EG2 and EG3, situated along the southern shores, correlate with substantial nitrogen contributions from agricultural landscapes. Likewise, terrestrial carbon also predominates the northern shores of the gulf. The northern coastline is characterized as the second most significant area in terms of TN and TP concentrations. In contrast to the southern region, the northern sector is characterized by denser forestation and a reduction in agricultural land, with the urban center of Erdek situated within this area. It is evident that minor stream systems traversing through the forests and agricultural regions flow into the gulf, thereby enriching these areas with carbon and nutrients. In the eastern sector of the gulf, organic carbon is determined to be autochthonous. The accumulation of BSi and CDP further substantiates that the southern

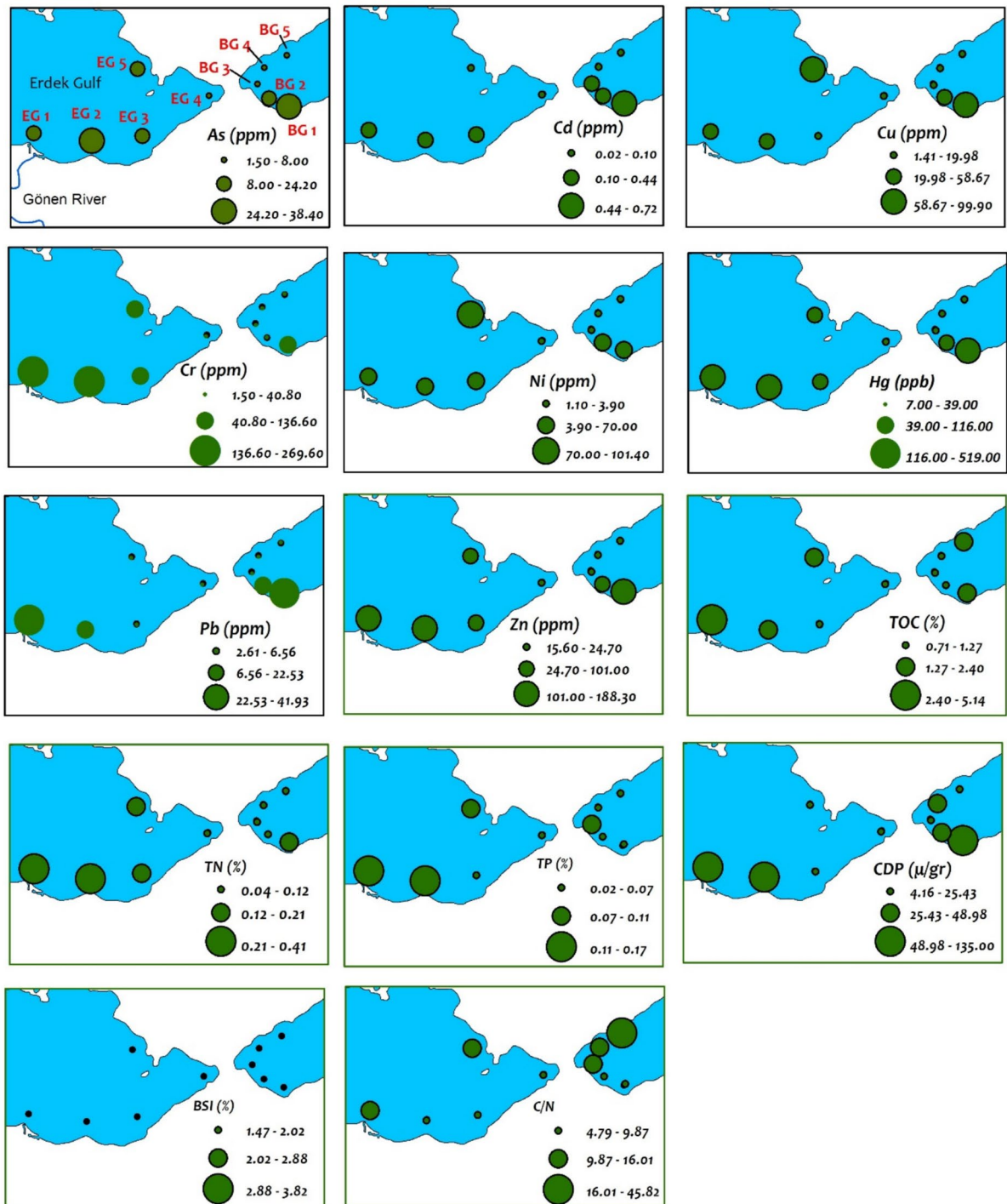


Fig. 5 Spatial distribution of PTEs and primary production proxies

shores of the gulf demonstrate heightened productivity in terms of primary production (Fig. 5).

In Bandırma Gulf, the concentration of TN exhibits its peak levels (0.13–0.21%) within the southern sector, whereas a downward trend (0.04–0.1%) is observed in the western and northern sections. It is hypothesized that urbanization serves as the primary contributor of nitrogen within Bandırma Gulf. The concentration of TP was recorded at its zenith (0.11%) in proximity to the coastline adjacent to the fertilizer manufacturing facility situated in the western region of the gulf. This is succeeded by the coastal area surrounding the urban center. Accumulation levels are comparatively diminished along the northern shores, which are characterized by agricultural lands, forested regions, and minor residential zones. The highest accumulation of TOC (1.83%) was identified in the northern segment of the gulf, followed by the southern and western sectors (1.74%, 1.27%, respectively). Elevated C/N ratios (14.16–45.82%) indicate that the substantial TOC accumulation in the northern area is attributed to the forested landscapes. A minor stream traverses the forested regions and discharges into the sampling locality within this area. In the southern coastal zone, recognized as the second most significant region for TOC accumulation, the carbon source is derived algal autochthonous blooms (C/N: 8.26–9.87). The peak values of CDP (82.37 µg/g) were also observed in this vicinity. A correlational relationship is evident between the heightened nitrogen levels and the proliferation of algae in this area. BSi exhibited a marked increase (2.68–2.89%) along the northern shores. In this locale, where the C/N ratio attained a value of 45 and the concentrations of TN and TP are relatively modest, it is conjectured that the accumulation of BSi may be of phytolith origin. The proximity of the station with the highest recorded value to a forested area, along with its influx from a stream, lends support to this hypothesis (Fig. 5).

3.4 Spatial Distribution and Accumulation Processes of PTEs

Elements known for their toxic properties to both ecosystems and human health (Cu, Pb, Zn, Ni, Cr, As, Cd, and Hg) were selected and used. The distribution of PTEs obtained from Bandırma Gulf are illustrated in Fig. 5. It is discerned that PTE accumulations are predominantly localized within the southern sector

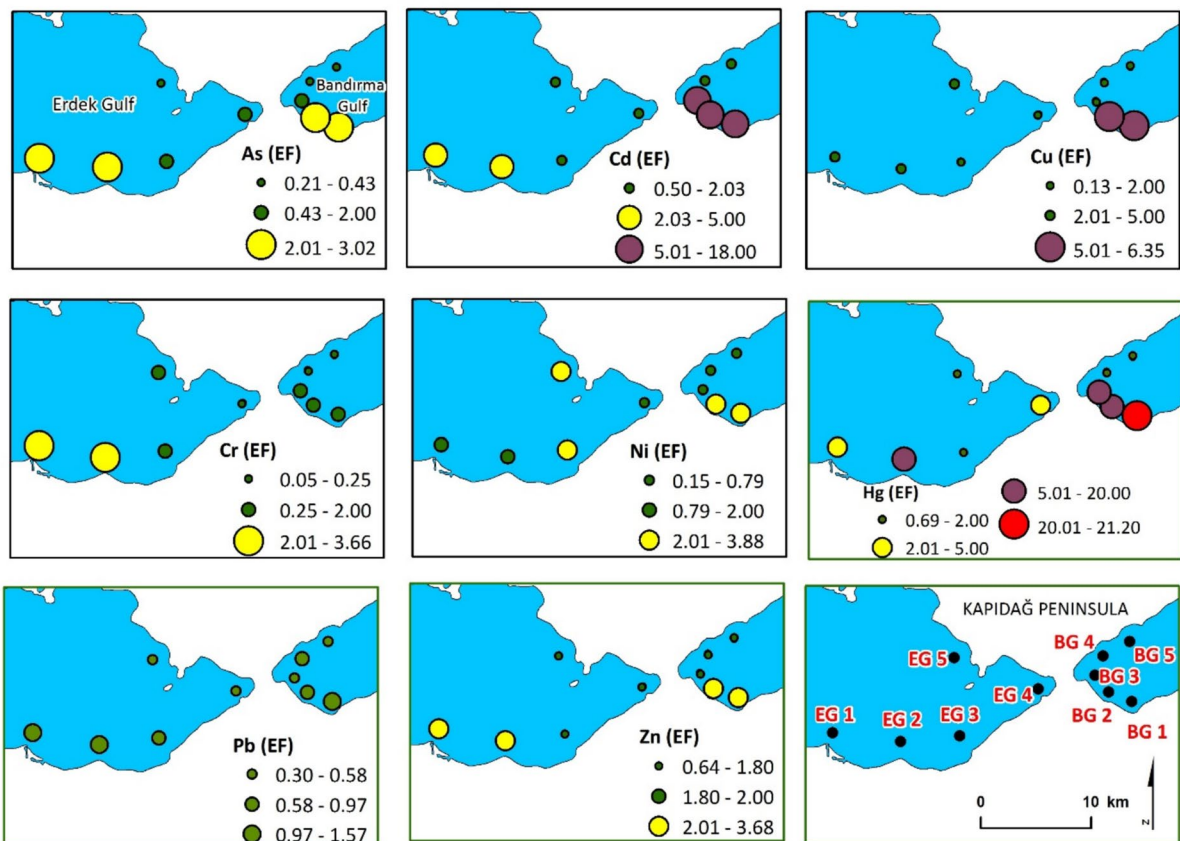
of Bandırma Gulf. The port area represents the locus where all elements are detected at their peak concentrations. It is postulated that the interplay between maritime traffic and the effluent discharges from prior years is accountable for this accumulation. In their investigation at Dafeng Port, Chen et al. (2020) recognized terrestrial transportation, port operations, and industrial establishments as the principal contributors to the enrichment of Pb, Cd, Zn, Cr, As, and Hg. Notably, the accumulation of Pb was correlated with port-related activities (Özkan et al., 2022). The values, which exhibit a decrement along the western coastline for all elements, persist in diminishing along the northern coastline for Cu, Ni, Cd, and Hg, whilst other elements demonstrate an increment in the northern region. In comparison to the results of a previous study conducted on the surface sediments of Bandırma Gulf, this research reveals that the maximum concentrations of Ni, Cu, and Zn are elevated, whereas the concentrations of Cr and Cd are diminished. The concentration of Pb was identified to be nearly equivalent (Mülayim et al., 2012).

The heavy metal concentrations detected in the Erdek and Bandırma Gulfs exhibit notable variations when compared to other referenced gulfs. Erdek Gulf shows particularly high levels of Cr (133.88 ppm), Zn (104.32 ppm), and Ni (42.92 ppm), with values comparable to or even exceeding those found in more industrialized areas such as the Gulf of Pozzuoli and the East China Sea. In Bandırma Gulf, relatively elevated levels of Cd (0.30 ppm) and Hg (0.108 ppb) suggest the influence of industrial activities. For instance, the Cd concentration in Bandırma is comparable to that in Pozzuoli (0.40 ppm) and Admiralty Bay (0.40 ppm). The Cr concentration in Erdek Gulf stands out as significantly higher than in all other compared regions. These findings indicate that both gulfs are subject to strong regional or anthropogenic influences, particularly in terms of specific PTE pollutants. For detailed comparative data, see the Table 3.

The EF values employed to ascertain the anthropogenic nature of the elements indicate that, with the exception of Cd and Cr, the maximum concentrations are situated in the southern region (Bandırma port area), whereas these two elements attain their peak levels along the western coastline. The values seemingly reach their nadir along the northern coastline. Along the southern coastal regions, significant

Table 3 Comparison of the average values obtained from the Erdek and Bandırma Gulfs with those reported for other gulfs world-wide

Location	Cu (ppm)	Pb (ppm)	Zn (ppm)	Ni (ppm)	As (ppm)	Cd (ppm)	Cr (ppm)	Hg (ppb)	References
Erdek Gulf	42.65	24.46	104.32	42.92	18.48	0.22	133.88	0.051	This study
Bandırma Gulf	33.01	13.75	56.52	25.66	12.78	0.30	26.90	0.108	This study
Persian Gulf	16.10	-	-		3.3	0.18	54.90	0.017	Rezaei et al., (2021)
Çandarlı Gulf	30.80	43.81	106.12	40.41	28.50	-	80.40		Duman et al., (2025)
Pozzuoli Gulf	15.10	115	179	6.30	22.4	0.40	15.50	0.7	Arienzo et al., (2020)
East China Sea	23.70	24.60	83.80	30.80	7.74	0.20	71.40	0.09	Zhang et al., (2020)

**Fig. 6** Spatial distribution of EF values

levels of accumulation were observed for Cu and Cd, whereas moderate accumulation was noted for Zn, Ni, and As. Within the port zone of this area, Hg attained very high levels of accumulation (Fig. 6). A research investigation conducted in a port indicated that Cd and Pb reached alarming levels in this vicinity,

alongside an observed enrichment in Cu and Zn elements (Lin et al., 2013). The accumulation observed in the interior regions of ports, characterized by restricted water movement and stagnant conditions, serves as a significant indicator of anthropogenic influences and sediment deposition (Radomirović

et al., 2023). Numerous studies have posited that the enrichment of Cu is linked to maritime activities (Ota et al., 2021). An alternative potential source of Cu within Bandırma Gulf could stem from emissions originating from the fertilizer manufacturing facility and atmospheric deposition (Radomirović et al., 2023).

In the western area of the gulf, significant accumulation was noted for Cd (18.02) and Hg (8.67), whereas the EF values for other elements indicated low levels of contamination. Along the northern coast, all elements were determined to be of natural origin. The region where Cd attained its peak levels is situated adjacent to the fertilizer production facility. Studies show that the rocks containing phosphate, which are crucial for synthetic phosphate fertilizer production, typically exhibit high concentrations of Cd (Sönmez et al., 2014). The observed increase in this area is believed to be attributable to the operational activities of the factory. Fertilizer plants are among the significant anthropogenic sources contributing to the pollution of air (Pandey, 2005), water (Noli et al., 2024), and soil. The fertilizer factories located in the Bandırma Bay area have been intensifying their production activities for over half a century. This ongoing industrial activity has led to an increased level of ecological risk in the region. Recent studies conducted on soils surrounding the fertilizer plants in Bandırma Bay indicate a marked enrichment of PTEs, particularly Cd and Cu (Parlak & Dengiz, 2025). These findings suggest that fertilizer plants pose a threat not only to the marine ecosystem but also to the terrestrial ecosystem, thereby reinforcing this research outcomes. Bandırma Port is a significant component of the Turkish Straits System, which experiences heavy maritime traffic. Shipping emissions not only increase the levels and alter the composition of particulate and gaseous pollutants but can also influence new particle formation processes in urban areas (Kuzu et al., 2021). Particulate matter released into the atmosphere eventually accumulates in the sediment through deposition processes. The pronounced accumulation of elements within the port zone is attributed to maritime traffic and the accumulation of urban waste that has been discharged into the region over an extended period. Under favorable circumstances, these materials could potentially be released back into the water, posing a notable risk to both marine ecosystems and public health. The radar

chart drawn according to the average of the EF values is presented in Fig. 7.

Igeo trend (Fig. 4b) is generally consistent with the findings of the EF (Fig. 4a). The elements Pb and Cr indicate minimal levels (−2.18 and −3.31 as average values respectively.) of accumulation throughout the gulf. Cu, Zn, Ni, and As are present within the uncontaminated to moderately contaminated spectrum in the southern zone, yet do not exhibit contamination in other areas. Cd and Hg, however, have surfaced as the most concerning metals. Cd demonstrates uncontaminated to strongly contamination, whereas Hg is recorded at extremely levels of accumulation. The radar chart drawn according to the average of the Igeo values is presented in Fig. 7.

The raw distributions of PTEs exhibit variations between the northern and southern shores of Erdek Gulf. Cu and Ni attain their peak concentrations along the northern shores, while Pb, Zn, As, Cd, Cr, and Hg are identified at their highest concentrations on the southern shores. The southern shores of the Gulf are predominantly characterized by agricultural land utilization. The Gönen River, which traverses agricultural regions, discharges into this area. The sampling site on the northern shore of the gulf is situated adjacent to the Erdek district center. In this locale, the cumulative effects of both residential and smaller agricultural areas, in contrast to the southern region, can be discerned (Fig. 5). When compared to outcomes from another surface sediment investigation conducted within the same vicinity, the maximum concentrations of Ni, Cu, Zn, and Cr observed in our study were found to exceed those previously recorded, while values for Cd and Pb were comparatively lower (Mülayim et al., 2012).

Based on the EF results utilized to ascertain the origins of the elements, the southern shore zone of the Erdek Gulf is classified as moderately contaminated with Cu, Zn, Ni, As, Cd, and Cr, exhibiting moderate to significant contamination with Hg. In the northern zone, moderate contamination is noted for Cu and Ni. Pb does not demonstrate anthropogenic enrichment across any segment of the gulf and is entirely lithogenic in its origin (Fig. 6). Mülayim et al., (2012) indicated that Cd, Pb, and Hg exhibited enrichment in both Erdek and Bandırma Gulfs. While the findings for Cd and Hg align with previous research, Pb enrichment was not identified in our investigation. The Igeo results, derived for analogous purposes,

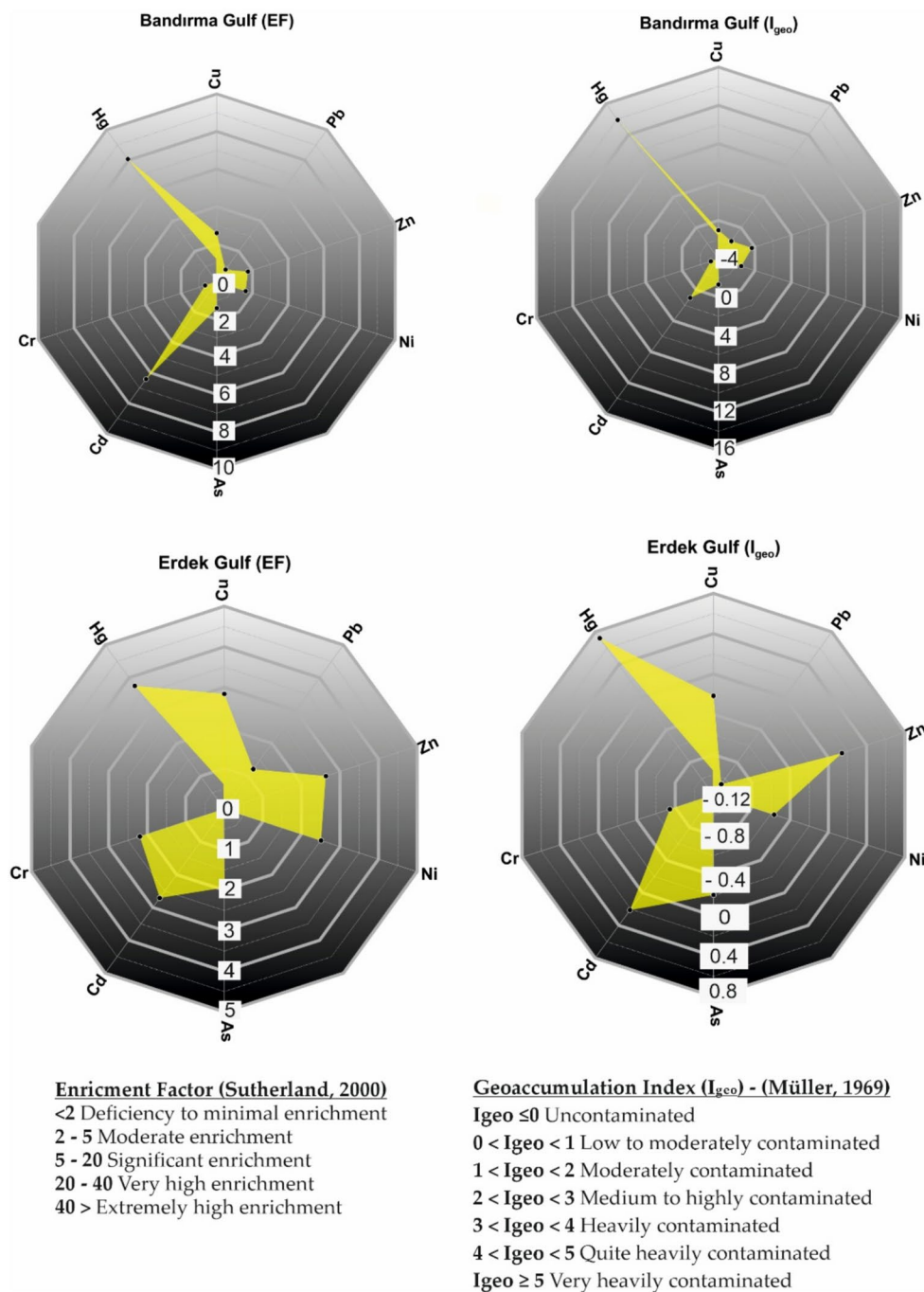


Fig. 7 Radar Graphs of EF and I_{geo} values

are congruent with EF values. The observed enrichment of Cu, Zn, Ni, As, and Cd along the southern shores of the gulf is hypothesized to be of agricultural origin (fertilizers, pesticides, herbicides, etc.)

(Aykir et al., 2023; Chen et al., 2020; Fural et al., 2022; Radomirović et al., 2023; Sönmez et al., 2014; Ustaoglu et al., 2020). Given that elevated TP concentrations were also detected in the same area, it can be

inferred that these pollutants are of agricultural origin. The Cu and Ni present in the northern zone may also derive from agricultural activities, albeit from a more limited area. Hg may stem from atmospheric deposition and the utilization of coal (Radomirović et al., 2023). The radar charts drawn according to the average of the I_{geo} and EF values is presented in Fig. 7.

In the analysis of the spatial distribution of ecological risks attributable to various elements, the southern region of Bandırma Gulf is prominently identified as the locus of concentrated risks. This is closely succeeded by the western coastal zone, whereas the northern coastline of the gulf is characterized by a minimization of such risks. Despite a marked increase in the concentrations of Cu, Pb, Zn, Ni, As, and Cr within the southern zone, the mRI values for these elements did not surpass the established threshold at any sampling location. It is, however, pertinent to note that Cu and As exhibited values approaching the threshold, particularly within the vicinity of the port area. Cd presents a potential ecological risk that falls within the significant to very high range in both the southern and western zones. The area deemed most vulnerable due to Cd contamination was identified adjacent to the fertilizer manufacturing facility. Hg has surfaced as the element associated with an exceptionally very high ecological risk within the southern and western sectors of the gulf. The port area registered the highest level of risk associated with Hg. Conversely, the distribution of Hg along the northern shores of the gulf is categorized as low to moderate. The cumulative potential ecological risks of PTEs are maintained at low levels along the northern shores of Bandırma Gulf, whereas they exhibit a range from significant to very high along the southern and western shores. The port area represents the zone with the highest aggregate risk, closely followed by the vicinity of the fertilizer factory, which presents an equivalent degree of risk (Fig. 8).

TRI values were observed to be substantial (TRI: 15.86) in the port area, while in BG2, situated on the southern coastline, a low risk was recorded (TRI: 9.46). The remaining monitoring stations reported values below the risk threshold (Fig. 9). The mean TRI value (5.67) throughout the Gulf suggests a low risk level. The primary contributors to the average TRI value are As (33%), Ni (20%), and Cu (12%),

succeeded in significance by Cr, Hg, Cd, Zn, and Pb, respectively.

The mRI presented by various elements within Erdek Gulf are predominantly concentrated in the southern sector. In this particular region, a significant potential ecological risk was assessed for Cd, alongside a high risk associated with Hg. It was ascertained that other elements exhibiting anthropogenic enrichment, such as Cu, Zn, Ni, Cr, and As, do not present a substantial risk, attributable to their comparatively lower accumulation levels and relative toxicological profiles. No discernible potential ecological risk was identified for any element along the northern coastline. Along the eastern coastline of the gulf, Hg was the sole element identified as posing a significant risk. The cumulative risk associated with the eight PTEs attains its zenith in the southern sector, succeeded by the eastern and northern coastlines. Risks that are considerable at the river mouth and its adjacent areas within the southern sector diminish to lower levels along the eastern section of the southern coastline. A moderate risk was recognized at the eastern shoreline (EG 4), whereas a low risk was observed along the northern shoreline. In light of this data, it can be inferred that the predominant source of pollution and associated risk in Erdek Gulf originates from the agricultural zones situated in the southern part of the Gulf, as well as the Gönen River, which transports contaminants from these regions into the Gulf (Fig. 8).

Calculations employing the TRI index, which utilizes an alternative methodological framework, suggest a moderate to significant risk for the southern region, a moderate risk for the northern region, and a low risk for the eastern region (Fig. 9). In the Erdek Gulf, TRI values were calculated to range between 1.04 and 16.53, with an average value of 9.83. The leading elements contributing to the TRI were marked as Cr at 28%, As at 27%, and Ni at 19%.

A noticeable discrepancy is observed between the TRI and mPER results. This difference arises from the fact that each index is based on distinct variables. The TRI is calculated using threshold values (TEL, PEL) derived from ecotoxicological studies conducted on certain organism groups in laboratory settings. In contrast, the mPER is calculated based on the degree of elemental enrichment in sediments. Including both indices, which rely on markedly different variables,

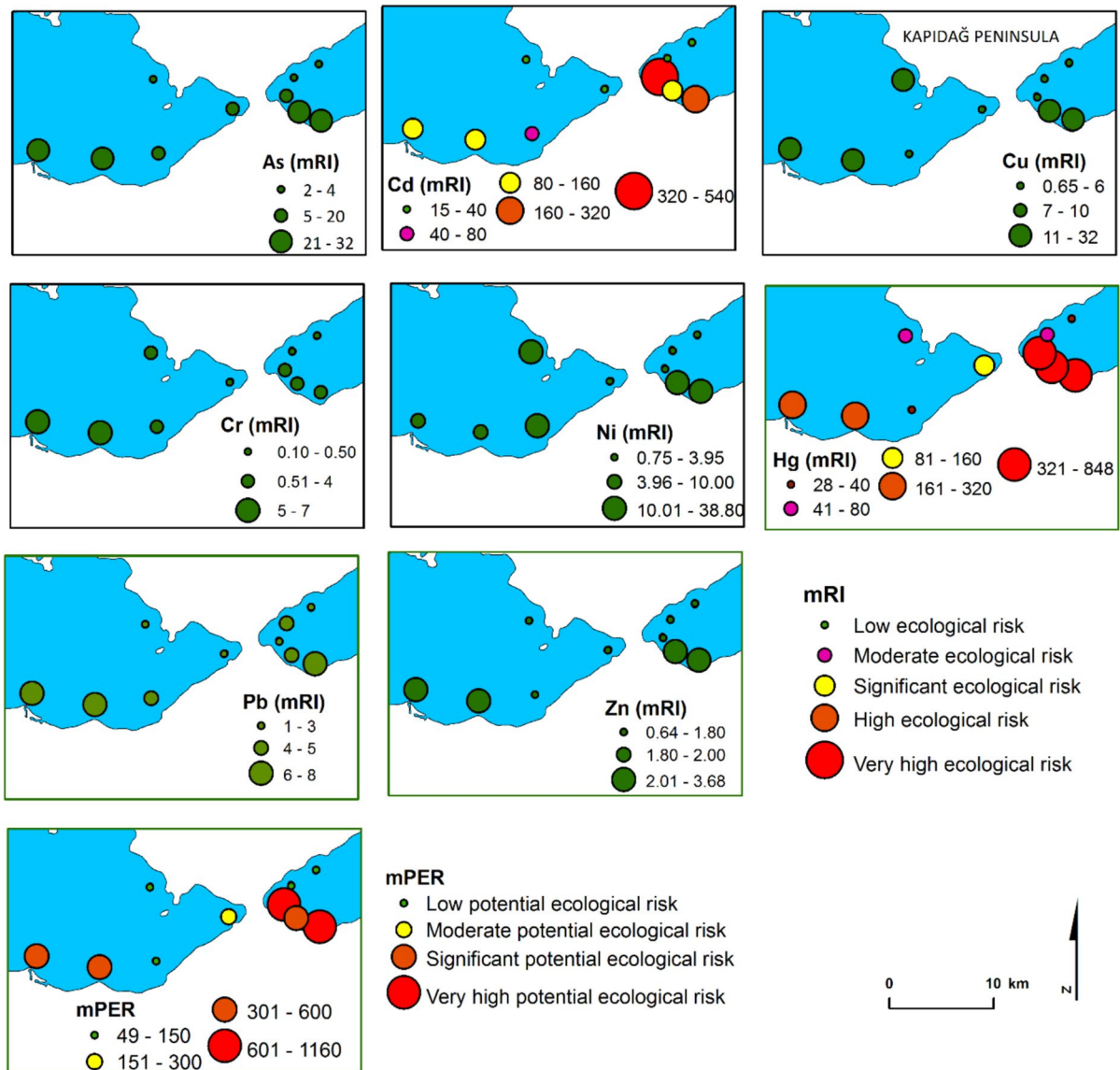


Fig. 8 Spatial distribution of ecological risk and potential ecological risk

enables a more comprehensive assessment and allows for evaluation from multiple perspectives.

3.5 Dinoflagellate Cyst Assemblages of Erdek and Bandırma Gulfs

This research serves as a preliminary study on modern dinoflagellate cysts in the Erdek and Bandırma Gulf region during a mucilage event. Sediments are generally characterized as fine sand and mud in the stations. A total of 15 dinoflagellate cyst taxa were

recognized in 5 stations in which sediments characterized as fine mud (Table 4). It is known that dinoflagellate cysts have similar hydrodynamic properties as fine silt particles (Dale et al., 1999). The seabed of other stations covered by sand that is unsuitable for dinoflagellate cyst deposition. Cyst of HAB species (Anderson et al., 1994; Balkis et al., 2016; Hallegraeff & Bolch, 1991; Ignatiades et al., 2007; Satake et al., 1997) were discovered in the surface sediments of the study area, including cyst of *Alexandrium* sp., *Operculodinium centrocarpum*, *Lingulodinium*

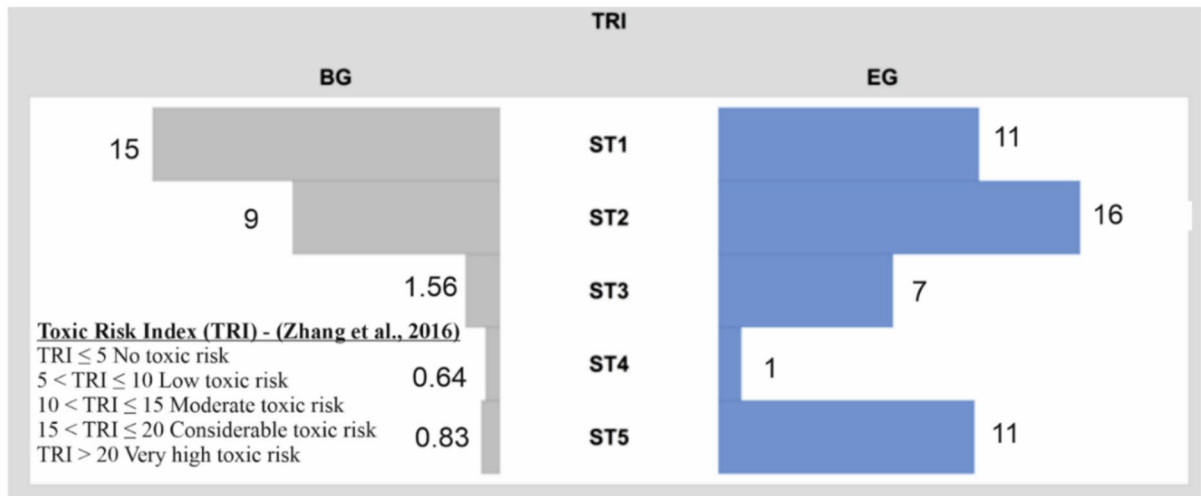


Fig. 9 Butterfly Chart of Toxic Risk Index (The abbreviation 'ST' in Figure refers to the sampling stations)

Table 4 Paleontological name of dinoflagellate cysts and their biological affinities according to references: Wall and Dale (1968), Head (1996), Rochon et al., (1999), Matsuoka and Fukuyo (2000), Radi et al. (2013), Matsuoka et al. (2009)

Paleontological name (cyst species)	Biological name (affinity)	EG1	EG2	EG3	EG4	BG1
Autotrophic species						
Cyst of <i>Alexandrium</i> sp.	<i>Alexandrium</i> sp.	+	+	+	+	+
<i>Lingulodinium machaerophorum</i>	<i>Lingulodinium polyedrum</i>	-	-	-	+	+
<i>Operculodinium centrocarpum</i>	<i>Protoceratium reticulatum</i>	-	-	-	+	+
Cyst of <i>Scrippsiella</i> spp.	<i>Scrippsiella</i> spp.	-	-	-	+	+
<i>Spiniferites bentorii</i>	<i>Gonyaulax digitalis</i>	+	+	+	+	+
<i>Spiniferites hyperacanthus</i>	<i>Gonyaulax spinifera</i> complex	+	+	+	+	+
<i>Spiniferites mirabilis</i>	<i>Gonyaulax spinifera</i> complex	+	+	+	+	+
<i>Spiniferites ramosus</i>	<i>Gonyaulax spinifera</i> complex	+	+	+	+	+
Heterotrophic species						
Cyst of <i>Polykrikos kofoidii</i>	<i>Polykrikos kofoidii</i>	-	-	-	+	+
Cyst of <i>Polykrikos schwartzii</i>	<i>Polykrikos schwartzii</i>	-	-	-	+	+
<i>Quinquecupis concreta</i>	<i>Protooperidinium leonis</i>	+	+	+	+	+
<i>Selenopemphix quanta</i>	<i>Protooperidinium conicum</i>	+	+	+	+	+
<i>Stelladinium stellatum</i>	<i>Protooperidinium stellatum</i>	+	+	+	+	+
Cyst of <i>Diplopsalis lenticula</i>	<i>Diplopsalis lenticula</i>	-	-	-	+	+
<i>Dubridinium caperatum</i>	<i>Preperidinium meunieri</i>	-	-	-	+	+
Total cyst number at stations		9	9	8	14	15

machaerophorum. The potentially toxic cyst species *Alexandrium* sp. and *Operculodinium centrocarpum*, previously reported in Marmara Sea (Aydin et al., 2015; Balkis et al., 2016) were also identified. It is understood that cysts of toxic species can contain more toxins than their motile vegetative cells

(Oshima et al., 1992), although the toxicity of certain species can vary with environmental conditions (Hamasaki et al., 2001; Röder et al., 2012). The cyst taxa recorded in Bandırma Gulf are similar to those found in the Aegean, Marmara, and southern Black Seas (Aydin & Uzar, 2014; Balkis et al., 2016;

Giannakourou et al., 2005; Koutsodendris et al., 2015; Mudie et al., 2004, 2011).

The Marmara Sea is characterized by high industrial pollution and sewage discharge, resulting in significant phytoplankton blooms (Mudie et al., 2017). Balkis et al., (2016) reported the presence dinoflagellates of *Alexandrium minutum*, *Alexandrium* sp., and *Heterocapsa triquetra* in Gemlik Gulf. Notable blooms of *Noctiluca scintillans* and mucilage formations, along with occurrences of *Gonyaulax fragilis*, have been observed in the Marmara Sea blooms (Mudie et al., 2017). Another study (Balkis-Ozdelice et al., 2020) identified various species: 61 from Dinophyceae, 28 from Bacillariophyceae, three from Dictyochophyceae, and one each from Euglenophyceae, Prasinophyceae, and Chlorophyceae classes in Erdek Gulf. Dinoflagellates dominated, with *Protoperidinium*, *Tripos*, and *Dinophysis* exhibiting the highest species counts. Additionally, two new species for Turkish seas were documented: *Gymnodinium dogieli* and *Heterodinium rigdeniae*. Cysts of non-toxic HAB dinoflagellates and potentially toxic species were found in sediment surveys of the Gemlik Gulf, albeit not in plankton samples (Balkis et al., 2016). These HABs have also been reported by (Aydin et al., 2015) in the Black Sea near the Bosphorus and Dardanelles Strait. Investigating dinoflagellate resting cysts in coastal areas is essential for elucidating factors driving algal growth and mucilage formations. Research on these cysts is critical for identifying local conditions associated with the increasing frequency of harmful algal blooms (HABs) in the Marmara Sea (Balkis et al., 2016). Analyzing dinoflagellate cyst distribution in sediments provides vital early warning indicators regarding toxic species and the likelihood of recurring HABs in designated regions (Mudie et al., 2017).

Dinoflagellate cysts located in sedimentary deposits are significant for analyzing the influence of eutrophication and environmental pollution on coastal regions. Hence, dinoflagellate cysts are considered vital bioindicators (Joyce et al., 2005). Paleontological research indicates that these cyst assemblages are instrumental in monitoring historical variations in surface water salinity in the Marmara and Black Seas Balkis et al., (2016). However, research on environmental factors and current dinoflagellate cysts in the Marmara Sea is sparse Balkis et al., (2016). Previous investigations predominantly focused on fossil

dinoflagellate cysts and organic-walled palynomorphs from sediment cores in the Marmara Sea Balkis et al., (2016). Conversely, studies of modern dinoflagellate cysts have mainly concentrated on Izmir Bay in the Aegean Sea (Aydin & Uzar, 2014; Aydin et al., 2011; Uzar et al., 2010).

Lingulodinium machaerophorum was the most observed species in the surface sediment of Bandırma Gulf. This species is commonly found in recent marine sediments from coastal and neritic environments and is an indicator of eutrophic waters (Dale et al., 1999; Zonneveld et al., 2013) or river influence in the Mediterranean region (Elshanawany et al., 2010). The autotrophic species *Lingulodinium polyedrum* thrives in areas with nutrient-rich surface waters and minimal turbulence, which hinders the growth of competing diatoms. *Lingulodinium polyedrum*, which are indicators of eutrophication depending on different regions, will be used in future studies to reveal the response of dinoflagellate cyst to eutrophication in Bandırma Gulf.

Operculodinium centrocarpum, as defined by (Wall & Dale, 1968), is linked to the motile stage *Protoceratium reticulatum* (Braarud, 1944). *Operculodinium centrocarpum* are cosmopolitan and can tolerate a wide range of temperatures and salinities (Rochon et al., 1999; Zonneveld et al., 2013); however, in the Marmara Sea, their populations appear most sensitive to summer temperatures (Balkis et al., 2016). In the Gulf of Naples, cyst production is primarily confined to the summer months (July–September) (Montresor et al., 1998). Conversely, in the Gulf of Gemlik, *O. centrocarpum* was found in all sediment samples collected in May 2012. Motile cells of this species were recorded in February 2005 in Kepez harbor on the Dardanelles Strait at the western entrance to the Marmara Sea (Turkoglu & Oner, 2010). *O. centrocarpum* is a yessotoxin-producing species that has caused mass mussel deaths on the west coast of South Africa (Horstman, 1981). Some strains of *P. reticulatum* from Spain also produce yessotoxin, though strains from eastern North America do not (Paz et al., 2007). It remains unclear whether Marmara Sea strains produce yessotoxins (Balkis et al., 2016).

Several studies attempted to establish the environmental indicator species were identified such as *Spiniferites mirabilis*, associated with warm waters; *O. centrocarpum*, linked to seasonally cold waters;

L. machaerophorum and *Brigantedinium* sp., indicative of low-salinity, eutrophic waters and red tide production; and *Selenopemphix quanta*, associated with trace elements like Zn and Pb (Dale et al., 1999; Marret & Zonneveld, 2003; Matsuoaka, 1999; Sangiorgi & Donders, 2004; Versteegh, 1994). Leading to the conclusion that dinoflagellate response to environmental variables may be spatially specific. As such, more studies are needed to add to the knowledge-base of dinoflagellate cysts as indicator of environmental conditions in our study area.

Scrippsiella trochoidea is a non-toxic bloom-forming HAB species common in both the Black Sea and Marmara Sea (Balkis et al., 2016; Gómez & Boicenco, 2004; Velikova et al., 1999) and is widespread in the Caspian Sea (Gogorev, 2006). Balkis et al., (2016) found cysts of *Scrippsiella precaria* in the Marmara Sea, along with cysts of *S. trochoidea* (up to 3200 cysts/cm³) and cysts of *S. trifida* (up to 2400 cysts/cm³), using untreated sediment samples. Additionally, Aydın et al., (2011) reported two hyaline-walled cysts of *Scrippsiella* sp. in Izmir Bay, the Aegean Sea, one of which may be *Scrippsiella trochoidea*. In our study, cysts of *Scrippsiella* sp. were found in untreated sediment samples in Bandırma Gulf, but further work is needed to identify the

species level. Due to insufficient sediment samples, dinoflagellate cyst abundance could not be determined. The dense layer formed by mucilage on the sediment made it very difficult to identify dinoflagellate cysts. Therefore, more species may be determined in future studies.

This research provides an overview of dinoflagellate cyst assemblages, including those of potentially HAB species, from toxic and harmful dinoflagellates widely distributed in surface sediments. This could pose a potential risk of HABs for future blooms, particularly in Bandırma Gulf. Future studies aim to correlate sediment pollution with dinoflagellate cyst abundance in this region.

3.6 Potential Sources of Variables

Hierarchical Cluster Analysis (HCA) identified three distinct clusters (Fig. 10). Cluster 1 comprises the anthropogenic metal and metalloid group (Cu, Pb, Ni, As). Within the upper section of the dendrogram, the very short distance between Cu and Pb indicates that these two metals originate from a common source or share similar transport mechanisms. Copper and lead are typically associated with automotive emissions, the paint industry, and

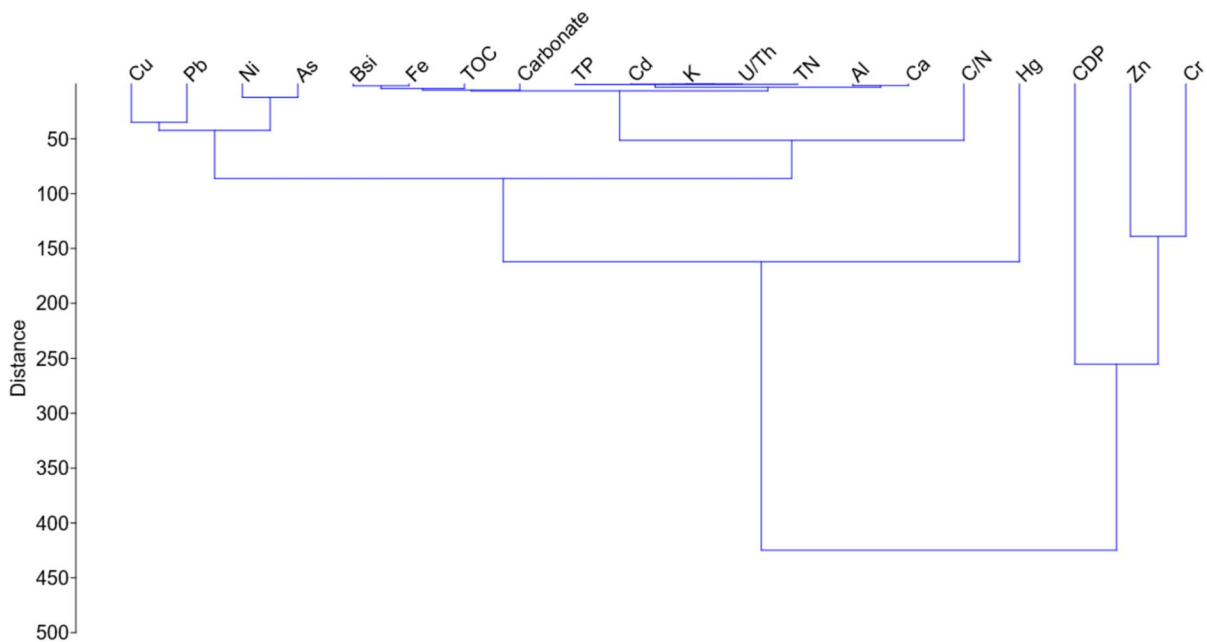


Fig. 10 Hierarchical clustering dendrogram of sediment core samples based on paleoenvironmental variables

urban surface runoff. While Enrichment Factor (EF) values for copper point to anthropogenic enrichment, it is observed that anthropogenic signals for Ni increase toward the core surface, despite not exceeding the critical threshold. This sub-group subsequently merges with Ni and As. The presence of these metals in the same cluster as urban pollutants like lead and copper suggests that industrial wastewater plays a dominant role in the distribution of these elements.

Cluster 2 is categorized as biogenic production and the sediment matrix, encompassing the variables BSi, Fe, TOC, Carbonate, TP, Cd, TN, Al, and Ca. This cluster constitutes the broadest and most complex portion of the dendrogram. Here, the strong correlations between BSi, TOC, Carbonate, TN, and TP reveal the geochemical framework of eutrophication processes within the gulf. Biogenic Silica (BSi) is situated on the same primary branch as lithogenic elements of terrestrial origin, such as Fe and Al. This supports the hypothesis that a significant portion of the BSi signal is transported by rivers via phytoliths found in terrestrial plants, rather than originating solely from marine diatoms. The total biogenic silica pool measured for paleoclimate and bioproductivity reconstructions is often mixed, incorporating both autochthonous aquatic remains (e.g., diatom frustules) and allochthonous phytoliths derived from terrestrial vegetation. (Bertrand et al., 2024). The inclusion of elements such as Fe and Al in this group indicates that organic matter is sequestered by clay minerals and iron oxides during sedimentation.

The proximity between TP and Cd is particularly noteworthy, reflecting the impact of fertilizers utilized in regional agricultural lands. Phosphate rocks naturally contain high concentrations of Cd, and these two elements enter the system concurrently during fertilizer production. The presence of cadmium within this biogenic cluster also demonstrates its tendency to be integrated into the marine food chain and to co-precipitate with organic matter. This group further integrates with TN, Al, and Ca. The presence of aluminum suggests that a fraction of the nitrogen may exist in an inorganic form (such as ammonium) bound to clay minerals, while the relationship between calcium and carbonate represents biogenic calcite production. Furthermore, the inclusion of the C/N ratio in this cluster confirms that the organic matter source is of marine origin.

Cluster 3 consists of isolated pollutants and chlorophyll degradation products (CDPs), comprising the variables Hg, CDP, Zn, and Cr. Located in the lower sections of the dendrogram, Hg and CDP join the main group at higher linkage distances compared to other variables. This indicates that the input mechanisms for these parameters are more distinct. Mercury, which exhibits the highest enrichment in the EF table, forms a relatively independent branch in the dendrogram. This suggests that Hg pollution, unlike other metals in the region, is controlled by different factors—potentially atmospheric deposition or a highly specific industrial point-source discharge. The relationship between CDP and Zn-Cr sheds light on "secondary pollution" mechanisms. CDP serves as a proxy for excessive algal blooms (such as mucilage). The association of zinc and chromium with this group suggests that these metals are either scavenged from the water column by microalgae and transported to the benthos or exhibit similar behavior during post-depositional diagenetic processes.

4 Conclusions

This study investigates the historical environmental changes and contemporary pollution levels in the Southern Marmara Sea. Historical proxies indicate low terrestrial erosion during AD 1241–1724, followed by increased erosion linked to rainfall, with recent terrestrial carbon inputs noted despite minimal erosion. Climatic changes influence material inputs and primary production processes. Over the last 200 years, urbanization and agriculture have significantly increased nitrogen and phosphorus inputs. This increase in nutrients has corresponded with heightened primary production in the same timeframe. The rise in primary production has decreased bottom water oxygen levels due to organic matter degradation, although recent intense rainfall has initiated a reversal of this trend. The surface distribution of primary production proxies indicates anthropogenic nutrient inputs as a critical concern. Major nitrogen sources include the Gönen River and urbanization in Bandırma Gulf. GIS mapping reveals the Gönen Stream in Erdek Gulf and a fertilizer facility in Bandırma Gulf contribute to phosphorus enrichment. Similar trends are noted in elemental pollution, with core samples showing enrichment over the last

200 years. mRI results suggest that, apart from Hg and Cd, other elements have not reached ecosystem-threatening levels. Hg and Cd are highlighted as elements necessitating urgent attention. Surface distribution maps identify the southern and western shores of Bandırma Gulf as high-risk for Cd, and the southern shore for Hg. In Erdek Gulf, southern shore areas exhibit significantly elevated risks, particularly with regard to Cd and Hg, with Hg also posing risks in the eastern section of the Gulf.

Based on these findings, the following recommendations are proposed:

The Gönen Stream, which flows through agricultural lands into Erdek Gulf, acts as a pollutant. The 12th article of the Mucilage Action Plan, which addresses combating pollution in internal basins, should be urgently implemented for the Gönen Stream Basin. Anthropogenic activities, especially agriculture and industry, should be brought under control.

A buffer channel should be constructed on both the right and left banks to weaken the connection between agricultural lands and the river in the Gönen Stream Basin. Water from agricultural lands accumulated in these channels should be treated before being discharged into the Gulf.

Anthropogenic activities in the southern and western parts of Bandırma Gulf should be effectively monitored. Waste inspections for ships anchoring in Bandırma Port should be intensified. To prevent the escalation of ecological risks associated with mucilage and PTEs in Bandırma and Erdek Gulfs, the most effective approach is to reduce "waste loads". The most effective way to reduce waste loads is to prevent the discharge of wastewater into the Gulf that has not passed through advanced biological treatment systems.

This study was conducted in the coastal zones of the Bandırma and Erdek Gulfs. To obtain more comprehensive information on the spatial distribution of elemental pollution within the gulfs, future studies may expand the sampling area and produce more detailed GIS-based maps. The study presents the first records of dinoflagellate cysts identified in the sediments of both gulfs. However, due to the high sand content at some of the sampling stations, the number of cysts was insufficient for abundance calculations; therefore, only a species list was compiled. Future research focusing on clay-rich sediments within the

gulfs would allow for the estimation of species abundances and facilitate the investigation of their relationships with chemical parameters.

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Data Availability The datasets generated and/or analyzed during the current study are not publicly available due to institutional data-sharing policies but are available from the corresponding author on reasonable request.

Declarations

Ethical Approval All authors have read, understood, and have complied as applicable with the statement on "Ethical responsibilities of Authors" as found in the Instructions for Authors.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process During the preparation of this work the author(s) used Chatgpt and Sci-Space in order to improve language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Conflict of interest The authors declare no competing interests.

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References

- Anderson, D. M., Kulis, D. M., Doucette, G. J., Gallagher, J. C., & Balech, E. (1994). Biogeography of toxic dinoflagellates in the genus *Alexandrium* from the northeastern United States and Canada. *Marine Biology*, 120(3), 467–478. <https://doi.org/10.1007/BF00680222>
- Arienzo, M., Ferrara, L., Toscanesi, M., Giarra, A., Donadio, C., & Trifuoggi, M. (2020). Sediment contamination by heavy metals and ecological risk assessment: The case of Gulf of Pozzuoli, Naples. *Italy. Marine Pollution Bulletin*, 155, Article 111149. <https://doi.org/10.1016/j.marpolbul.2020.111149>
- Atkinson, C. A., Jolley, D. F., & Simpson, S. L. (2007). Effect of overlying water pH, dissolved oxygen, salinity and sediment disturbances on metal release and sequestration from metal contaminated marine sediments. *Chemosphere*, 69(9), 1428–1437. <https://doi.org/10.1016/j.chemosphere.2007.04.068>
- Aydın, H., Matsuoka, K., & Minareci, E. (2011). Distribution of dinoflagellate cysts in recent sediments from Izmir Bay (Aegean Sea, Eastern Mediterranean). *Marine Micropaleontology*, 80(1), 44–52. <https://doi.org/10.1016/j.marmicro.2011.03.004>
- Aydın, H., & Uzar, S. (2014). Distribution and abundance of modern dinoflagellate cysts from Marmara, Aegean and Eastern Seas of Turkey. *Journal of Environmental Biology*, 35(2), 413–419.
- Aydın, H., Balci, M., Uzar, S., & Balkis, N. (2015). Dinoflagellate cyst assemblages in surface sediments of southwestern Black Sea and Çanakkale Strait (Dardanelles). *Frese-nius Environmental Bulletin*, 24(C12), 4789–4798.
- Aykir, D., Fural, Ş, Kükre, S., & Mutlu, Y. E. (2023). Element-based ecological and human health risk assessment in a lagoon system in a densely populated basin. *Oceanological and Hydrobiological Studies*, 52(1), 1–19. <https://doi.org/10.26881/oahs-2023.1.01>
- Badawy, W. M., Dului, O. G., El Samman, H., El-Taher, A., & Frontasyeva, M. V. (2021). A review of major and trace elements in Nile River and Western Red Sea sediments: An approach of geochemistry, pollution, and associated hazards. *Applied Radiation and Isotopes*, 170, Article 109595. <https://doi.org/10.1016/j.apradiso.2021.109595>
- Balkis, N., Balci, M., Giannakourou, A., Venetsanopoulou, A., & Mudie, P. (2016). Dinoflagellate resting cysts in recent marine sediments from the Gulf of Gemlik (Marmara Sea, Turkey) and seasonal harmful algal blooms. *Phycologia*, 55(2), 187–209. <https://doi.org/10.2216/15-93.1>
- Balkis-Ozdelice, N., Durmuş, T., Toklu-Alicli, B., & Balci, M. (2020). Phytoplankton composition related to the environmental conditions in the coastal waters of the Gulf of Erdek. *Indian Journal of Geo-Marine Sciences*, 49(09), 1545–1559.
- Bertrand, S., Tjallingii, R., Kylander, M. E., Wilhelm, B., Roberts, S. J., Arnaud, F., Brown, E., & Bindler, R. (2024). Inorganic geochemistry of lake sediments: A review of analytical techniques and guidelines for data interpretation. *Earth-Science Reviews*, 249, Article 104639. <https://doi.org/10.1016/j.earscirev.2023.104639>
- Biltekin, D., Eriş, K. K., Çağatay, M. N., Henry, P., & Yakupoğlu, N. (2023). New records of vegetation and climate changes in the Sea of Marmara during the Marine Isotope Stages 3, 4 and 5 (a-c). *Quaternary International*, 667, 1–18. <https://doi.org/10.1016/j.quaint.2023.06.005>
- Biltekin, D., Eriş, K. K., Schachner, A., Yakupoğlu, N., & Yakupoğlu, C. (2025). Late Holocene vegetation dynamics and climate variations through pollen analysis of sediments from Lake Sülük (Çorum, Türkiye). *Review of Palaeobotany and Palynology*, 334, Article 105280. <https://doi.org/10.1016/j.revpalbo.2025.105280>
- Blaauw, M. (2010). Methods and code for 'classical' age-modelling of radiocarbon sequences. *Quaternary Geochronology*, 5(5), 512–518. <https://doi.org/10.1016/j.quageo.2010.01.002>
- Braarud, T. (1944). *Morphological observations on marine dinoflagellate cultures* (Porella perforata, Gonialax tamarensis, Protoceratium reticulatum) (Avhandling utgitt av Det Norske Videnskaps-Akademi i Oslo. I. Mat.-Naturv. Klasse, No. 11). Jacob Dybwad.
- Brady, J. P., Ayoko, G. A., Martens, W. N., & Goonetilleke, A. (2015). Development of a hybrid pollution index for heavy metals in marine and estuarine sediments. *Environmental Monitoring and Assessment*, 187(5), 306. <https://doi.org/10.1007/s10661-015-4563-x>
- Çağatay, M. N., Erel, L., Bellucci, L. G., Polonia, A., Gasperini, L., Eriş, K. K., Sancar, Ü., Biltekin, D., Uçarkuş, G., Ülgen, U. B., & Damcı, E. (2012). Sedimentary earthquake records in the İzmit Gulf, Sea of Marmara, Turkey. *Sedimentary Geology, The 2011 Tohoku-Oki Tsunami*, 282, 347–359. <https://doi.org/10.1016/j.sedgeo.2012.10.001>
- Chen, Y., Liu, Q., Xu, M., & Wang, Z. (2020). Inter-annual variability of heavy metals pollution in surface sediments of Jiangsu coastal region, China: Case study of the Dafeng Port. *Marine Pollution Bulletin*, 150, Article 110720. <https://doi.org/10.1016/j.marpolbul.2019.110720>
- Choudhary, S., Tiwari, A. K., Nayak, G. N., & Bejugam, P. (2018). Sedimentological and geochemical investigations to understand source of sediments and processes of recent past in Schirmacher Oasis, East Antarctica. *Polar Science*, 15, 87–98. <https://doi.org/10.1016/j.polar.2018.01.003>
- Costantini, M. L., Agah, H., Fiorentino, F., Irandoost, F., Trujillo, F. J. L., Careddu, G., Calizza, E., & Rossi, L. (2021). Nitrogen and metal pollution in the southern Caspian Sea: A multiple approach to bioassessment. *Environmental Science and Pollution Research*, 28(8), 9898–9912. <https://doi.org/10.1007/s11356-020-11243-8>
- Dale, B., Thorsen, T. A., & Fjellsa, A. (1999). Dinoflagellate cysts as indicators of cultural eutrophication in the

- Oslofjord, Norway. *Estuarine, Coastal and Shelf Science*, 48(3), 371–382. <https://doi.org/10.1006/ecss.1999.0427>
- Dan, S. F., Li, S., Yang, B., Cui, D., Ning, Z., Huang, H., Zhou, J., & Yang, J. (2021). Influence of sedimentary organic matter sources on the distribution characteristics and preservation status of organic carbon, nitrogen, phosphorus, and biogenic silica in the Daya Bay, northern South China Sea. *Science of the Total Environment*, 783, Article 146899. <https://doi.org/10.1016/j.scitotenv.2021.146899>
- DeMaster, D. J. (1981). The supply and accumulation of silica in the marine environment. *Geochimica Et Cosmochimica Acta*, 45(10), 1715–1732. [https://doi.org/10.1016/0016-7037\(81\)90006-5](https://doi.org/10.1016/0016-7037(81)90006-5)
- Duman, M., Eronat, A. H., Talas, E., Gerdanlı, M. C., & Tükel, C. (2025). Understanding heavy metal pollution dynamics in Çandarlı Gulf, western Türkiye: Insights from spatial analysis, sediment transport modeling and ecological risk assessment. *Continental Shelf Research*, 285, Article 105393. <https://doi.org/10.1016/j.csr.2024.105393>
- Elshanawany, R., Zonneveld, K., Ibrahim, M. I., & Kholeif, S. E. A. (2010). Distribution patterns of recent organic-walled dinoflagellate cysts in relation to environmental parameters in the Mediterranean Sea. *Palynology*, 34(2), 233–260. <https://doi.org/10.1080/01916121003711665>
- Erginal, A., Cagatay, M., Selim, H., Karabiyikoglu, M., Çakir, Ç., Yakupoğlu, N., Acar, D., Akbaş, A., & Kaya, H. (2019). Multi-proxy sedimentary records of dry-wet climate cycles during the last 2 ka from Lake Çıldır, east Anatolian Plateau, Turkey. *Geografia Fisica e Dinamica Quaternaria*. <https://doi.org/10.4461/gfdq.2019.42.5>
- Fritz, K. M., Pond, G. J., Johnson, B. R., & Barton, C. D. (2019). Coarse particulate organic matter dynamics in ephemeral tributaries of a Central Appalachian stream network. *Ecosphere*, 10(3), Article e02654. <https://doi.org/10.1002/ecs2.2654>
- Fural, Ş., Kükrer, S., & Cürebal, İ. (2020). Geographical information systems based ecological risk analysis of metal accumulation in sediments of İkizcetepeler Dam Lake (Turkey). *Ecological Indicators*, 119, Article 106784. <https://doi.org/10.1016/j.ecolind.2020.106784>
- Fural, Ş., Kükrer, S., Cürebal, İ., & Aykır, D. (2022). Ecological degradation and non-carcinogenic health risks of potential toxic elements: A GIS-based spatial analysis for Doğancı Dam (Turkey). *Environmental Monitoring and Assessment*, 194(4), 269. <https://doi.org/10.1007/s10661-022-09870-4>
- Gaudette, H. E., Flight, W. R., Toner, L., & Folger, D. W. (1974). An inexpensive titration method for the determination of organic carbon in recent sediments. *Journal of Sedimentary Research*, 44(1), 249–253. <https://doi.org/10.1306/74D729D7-2B21-11D7-8648000102C1865D>
- Giannakourou, A., Orlova, T. Y., Assimakopoulou, G., & Pagou, K. (2005). Dinoflagellate cysts in recent marine sediments from Thermaikos Gulf, Greece: Effects of resuspension events on vertical cyst distribution. *Continental Shelf Research, Impact of Natural and Trawling Events on Resuspension, Dispersion and Fate of Pollutants (INTERPOL)*, 25(19), 2585–2596.
- Gogorev, R. (2006). *Caspian Sea Biodiversity Project—Checklist for Caspian Sea phytoplankton*. https://www.zin.ru/projects/caspidiv/caspian_phytoplankton.html
- Gómez, F., & Boicenco, L. (2004). An annotated checklist of dinoflagellates in the Black Sea. *Hydrobiologia*, 517(1), 43–59. <https://doi.org/10.1023/B:HYDR.0000027336.05452.07>
- Hakanson, L. (1980). An ecological risk index for aquatic pollution control. A sedimentological approach. *Water Research*, 14(8), 975–1001. [https://doi.org/10.1016/0043-1354\(80\)90143-8](https://doi.org/10.1016/0043-1354(80)90143-8)
- Hallegræff, G. M., & Bolch, C. J. (1991). Transport of toxic dinoflagellate cysts via ships' ballast water. *Marine Pollution Bulletin*, 22(1), 27–30. [https://doi.org/10.1016/0025-326X\(91\)90441-T](https://doi.org/10.1016/0025-326X(91)90441-T)
- Hamasaki, K., Horie, M., Tokimitsu, S., Toda, T., & Taguchi, S. (2001). Variability in toxicity of the dinoflagellate *Alexandrium tamarense* isolated from Hiroshima Bay, Western Japan, as a reflection of changing environmental conditions. *Journal of Plankton Research*, 23(3), 271–278. <https://doi.org/10.1093/plankt/23.3.271>
- Harrison, P. J., Conway, H. L., Holmes, R. W., & Davis, C. O. (1977). Marine diatoms grown in chemostats under silicate or ammonium limitation. III. Cellular chemical composition and morphology of *Chaetoceros debilis*, *Skeletonema costatum*, and *Thalassiosira gravida*. *Marine Biology*, 43(1), 19–31. <https://doi.org/10.1007/BF00392568>
- Head, M. J. (1996). Modern dinoflagellate cysts and their biological affinities. In J. Jansoni & D. C. McGregor (Eds.), *Palynology: principles and applications* (vol. 3, pp. 1197–1248). American Association of Stratigraphic Palynologists Foundation.
- Horstman, D. A. (1981). Reported red-water outbreaks and their effects on fauna on the west and south coast of South Africa, 1959–1980. *Fishery Bulletin, Sea Fisheries Institute, South Africa*, 15, 71–88.
- Ignatiades, L., Gotsis-Skretas, O., & Metaxatos, A. (2007). Field and culture studies on the ecophysiology of the toxic dinoflagellate *Alexandrium minutum* (Halim) present in Greek coastal waters. *Harmful Algae*, 6(2), 153–165. <https://doi.org/10.1016/j.hal.2006.04.002>
- Jenny, J.-P., Normandeau, A., Francus, P., Taranu, Z. E., Gregory-Eaves, I., Lapointe, F., Jautzy, J., Ojala, A. E. K., Dorioz, J.-M., Schimmelmänn, A., & Zolitschka, B. (2016). Urban point sources of nutrients were the leading cause for the historical spread of hypoxia across European lakes. *Proceedings of the National Academy of Sciences of the United States of America*, 113(45), 12655–12660. <https://doi.org/10.1073/pnas.1605480113>
- Joyce, L. B., Pitcher, G. C., du Randt, A., & Monteiro, P. M. S. (2005). Dinoflagellate cysts from surface sediments of Saldanha Bay, South Africa: An indication of the potential risk of harmful algal blooms. *Harmful Algae*, 4(2), 309–318. <https://doi.org/10.1016/j.hal.2004.08.001>
- Kam, E., & Önce, M. (2016). Pollution potential of heavy metals in the current sea sediments between Bandırma (Balıkesir) and Lapseki (Çanakkale) in the Marmara Sea. *Journal of Engineering Technology and Applied Sciences*, 1(3), Article Article 3. <https://doi.org/10.30931/jetas.297619>

- Koutsodendris, A., Brauer, A., Zacharias, I., Putyrskaya, V., Klemt, E., Sangiorgi, F., & Pross, J. (2015). Ecosystem response to human- and climate-induced environmental stress on an anoxic coastal lagoon (Etoliko, Greece) since 1930 AD. *Journal of Paleolimnology*, 53(3), 255–270. <https://doi.org/10.1007/s10933-014-9823-1>
- Kükrer, S., Erginal, A. E., Kılıç, Ş., Bay, Ö., Akarsu, T., & Öztura, E. (2020). Ecological risk assessment of surface sediments of Çardak Lagoon along a human disturbance gradient. *Environmental Monitoring and Assessment*, 192(6), 359. <https://doi.org/10.1007/s10661-020-08336-9>
- Kükrer, S., Fural, Ş., & Aykir, D. (2021). Aktaş gölü (ardahan) çökellerinde birincil paleo-üretim göstergelerinin zamansal değişiminin incelenmesi. *Ege Coğrafya Dergisi*, 30(2), Article 2. <https://doi.org/10.51800/ecd.992726>
- Kuzu, S. L., Bilgili, L., & Kiliç, A. (2021). Estimation and dispersion analysis of shipping emissions in Bandırma Port, Turkey. *Environment, Development and Sustainability*, 23(7), 10288–10308. <https://doi.org/10.1007/s10668-020-01057-6>
- Lin, Y.-C., Chang-Chien, G.-P., Chiang, P.-C., Chen, W.-H., & Lin, Y.-C. (2013). Multivariate analysis of heavy metal contaminations in seawater and sediments from a heavily industrialized harbor in Southern Taiwan. *Marine Pollution Bulletin*, 76(1), 266–275. <https://doi.org/10.1016/j.marpolbul.2013.08.027>
- Liu, J., Ye, S., Allen Laws, E., Xue, C., Yuan, H., Ding, X., Zhao, G., Yang, S., He, L., Wang, J., Pei, S., Wang, Y., & Lu, Q. (2017). Sedimentary environment evolution and biogenic silica records over 33,000 years in the Liaohe delta. *China. Limnology and Oceanography*, 62(2), 474–489. <https://doi.org/10.1002/lno.10435>
- Lorenzen, C. J. (1971). Chlorophyll-degradation products in sediments of Black Sea. *Woods Hole Oceanographic Institution Contribution*, 28, 426–428.
- Macdonald, D. D., Carr, R. S., & Calder, F. D. (1997). Development and evaluation of sediment quality guidelines for Florida coastal waters. *Oceanographic Literature Review*, 6(44), 638.
- MacDonald, D. D., Ingersoll, C. G., & Berger, T. A. (2000). Development and evaluation of consensus-based sediment quality guidelines for freshwater ecosystems. *Archives of Environmental Contamination and Toxicology*, 39(1), 20–31. <https://doi.org/10.1007/s002440010075>
- Makaroğlu, Ö., Küçükdemirci, M., Karlioğlu, N., Acar, D., Gürel, A., Dağdeviren, R., Yakupoğlu, N., Sabuncu, A., Altun, D., Karaöz, M., & Çağatay, M. (2022). Holocene sedimentary history of South Danamandra Lake: A peatland in west of İstanbul, Çatalca Peninsula, NW Turkey. *Turkish Journal of Earth Sciences*, 31(4), 410–423. <https://doi.org/10.55730/1300-0985.1809>
- Marret, F., & Zonneveld, K. A. F. (2003). Atlas of modern organic-walled dinoflagellate cyst distribution. *Review of Palaeobotany and Palynology*, 125(1), 1–200. [https://doi.org/10.1016/S0034-6667\(02\)00229-4](https://doi.org/10.1016/S0034-6667(02)00229-4)
- Matsuoka, K. (1999). Eutrophication process recorded in dinoflagellate cyst assemblages—A case of Yokohama Port, Tokyo Bay, Japan. *Science of the Total Environment*, 231(1), 17–35. [https://doi.org/10.1016/S0048-9697\(99\)00087-X](https://doi.org/10.1016/S0048-9697(99)00087-X)
- Matsuoka, K., & Fukuyo, Y. (2000). *Technical guide for modern dinoflagellate cyst study* (pp. 1–29). WESTPAC-HAB, Japan Society for the Promotion of Science.
- Matsuoka, K., Kawami, H., Nagai, S., Iwataki, M., & Takayama, H. (2009). Re-examination of cyst–motile relationships of Polykrikos kofoidii Chatton and Polykrikos schwartzii Bütschli (Gymnodinales, Dinophyceae). *Review of Palaeobotany and Palynology*, 154(1–4), 79–90. <https://doi.org/10.1016/j.revpalbo.2008.12.013>
- Meydan, A. F., Kalugin, I. A., Darin, A. V., Babich, V. V., Markovich, T. I., Rogozin, D. Y., Çağatay, M. N., Gülyüz, E., & Akkol, S. (2024). Paleoenvironmental reconstruction of Lake Van and Lake Erçek over the last millennium using varved sediments (Eastern Türkiye). *Quaternary International*, 714, Article 109581. <https://doi.org/10.1016/j.quaint.2024.10.013>
- Montresor, M., Zingone, A., & Sarno, D. (1998). Dinoflagellate cyst production at a coastal Mediterranean site. *Journal of Plankton Research*, 20(12), 2291–2312. <https://doi.org/10.1093/plankt/20.12.2291>
- Mudie, P. J., Rochon, A., Aksu, A. E., & Gillespie, H. (2004). Late glacial, Holocene and modern dinoflagellate cyst assemblages in the Aegean–Marmara–Black Sea corridor: Statistical analysis and re-interpretation of the early Holocene Noah’s Flood hypothesis. *Review of Palaeobotany and Palynology, Middle Latitude Dinoflagellates and Their Cysts*, 128(1), 143–167. [https://doi.org/10.1016/S0034-6667\(03\)00117-9](https://doi.org/10.1016/S0034-6667(03)00117-9)
- Mudie, P. J., Marret, F., Mertens, K. N., Shumilovskikh, L., & Leroy, S. A. G. (2017). Atlas of modern dinoflagellate cyst distributions in the Black Sea Corridor: From Aegean to Aral Seas, including Marmara, Black, Azov and Caspian Seas. *Marine Micropaleontology*, 134, 1–152. <https://doi.org/10.1016/j.marmicro.2017.05.004>
- Mudie, P. J., Leroy, S. A. G., Marret, F., Gerasimenko, N. P., Kholeif, S. E. A., Sapelko, T., & Filipova-Marinova, M. (2011). Nonpollen palynomorphs: Indicators of salinity and environmental change in the Caspian–Black Sea–Mediterranean corridor. In I. V. Buynevich, V. Yanko-Hombach, A. S. Gilbert, & R. E. Martin (Eds.), *Geology and Geoarchaeology of the Black Sea Region: Beyond the Flood Hypothesis* (p. 0). Geological Society of America. [https://doi.org/10.1130/2011.2473\(07\)](https://doi.org/10.1130/2011.2473(07))
- Mülayim, A., Balkis, N., Balkis, H., & Aksu, A. (2012). Distributions of total metals in the surface sediments of the Bandırma and Erdek Gulfs, Marmara Sea, Turkey. *Toxicological & Environmental Chemistry*, 94(1), 56–69. <https://doi.org/10.1080/02772248.2011.633914>
- Müller, G. (1969). Index of geoaccumulation in sediments of the Rhine River. *GeoJournal*, 2(3), 108–118.
- Noli, F., Sidirelli, M., & Tsamos, P. (2024). The impact of phosphate fertilizer factory on the chemical and radiological pollution of the surrounding marine area (seawater and sediments) in northwestern Greece. *Regional Studies in Marine Science*, 73, Article 103458. <https://doi.org/10.1016/j.rsma.2024.103458>
- Orhan, H., Delikan, A., Demir, A., Kapan, S., Olgun, K., Özmen, A., Sayın, Ü., Ekici, G., Aydın, H., Engin, B., & Tapramaz, R. (2021). Late Quaternary paleoclimatic and

- paleoenvironmental changes in the Konya Closed Basin (Konya, Turkey) recorded by geochemical proxies from lacustrine sediments. *Arabian Journal of Geosciences*, 14(9), 766. <https://doi.org/10.1007/s12517-021-07030-5>
- Oshima, Y., Bolch, C. J., & Hallegraeff, G. M. (1992). Toxin composition of resting cysts of *Alexandrium tamarense* (Dinophyceae). *Toxicon*, 30(12), 1539–1544. [https://doi.org/10.1016/0041-0101\(92\)90025-Z](https://doi.org/10.1016/0041-0101(92)90025-Z)
- Ota, Y., Suzuki, A., Yamaoka, K., Nagao, M., Tanaka, Y., Irizuki, T., Fujiwara, O., Yoshioka, K., Kawagata, S., Kawano, S., & Nishimura, O. (2021). Geochemical distribution of heavy metal elements and potential ecological risk assessment of Matsushima Bay sediments during 2012–2016. *Science of the Total Environment*, 751, Article 141825. <https://doi.org/10.1016/j.scitotenv.2020.141825>
- Özen, Ç., Sarı, E., Arslan Kaya, T. N., Gül, M., Kurt, M. A., & Adeel, M. (2023). Paleo environmental pollution assessment of Erdek and Bandırma Bays in the Sea of Marmara, Türkiye. *Soil and Sediment Contamination: An International Journal*, 0(0), 1. <https://doi.org/10.1080/15320383.2023.2200863>
- Özkan, E. Y., Fural, Ş, Kükrer, S., & Büyükkışık, H. B. (2022). Seasonal and spatial variations of ecological risk from potential toxic elements in the southern littoral zone of İzmir Inner Gulf, Turkey. *Environmental Science and Pollution Research*, 29(41), 62669–62689. <https://doi.org/10.1007/s11356-022-19987-1>
- Özşahin, E. (2011). Relationship between land use with geomorphological units in the Gonen Basin (Balıkesir). *Adıyaman Üniversitesi Sosyal Bilimler Enstitüsü Dergisi*, 7, 7. <https://doi.org/10.14520/adyusbd.115>
- Paerl, H. W. (2006). Assessing and managing nutrient-enhanced eutrophication in estuarine and coastal waters: Interactive effects of human and climatic perturbations. *Ecological Engineering, Advances in Coastal Habitat Restoration in the Northern Gulf of Mexico*, 26(1), 40–54. <https://doi.org/10.1016/j.ecoleng.2005.09.006>
- Pandey, J. (2005). Evaluation of air pollution phytotoxicity downwind of a phosphate fertilizer factory in India. *Environmental Monitoring and Assessment*, 100(1), 249–266. <https://doi.org/10.1007/s10661-005-6509-1>
- Parlak, M., & Dengiz, O. (2025). Assessment of soil contamination by potentially toxic elements around fertilizer factory: An ecological and health risks in Bandırma, NW Türkiye. *Environmental Forensics*, 0(0), 1–17. <https://doi.org/10.1080/15275922.2025.2490476>
- Paz, B., Riobó, P., Ramilo, I., & Franco, J. M. (2007). Yessotoxins profile in strains of *Protoceratium reticulatum* from Spain and USA. *Toxicon*, 50(1), 1–17. <https://doi.org/10.1016/j.toxicon.2007.02.005>
- Radi, T., Bonnet, S., Cormier, M.-A., de Vernal, A., Durantou, L., Faubert, E., Head, M. J., Henry, M., Pospelova, V., Rochon, A., & Van Nieuwenhove, N. (2013). Operational taxonomy and (Paleo-) autecology of round, brown, spiny dinoflagellate cysts from the Quaternary of high northern latitudes. *Marine Micropaleontology*, 98, 41–57. <https://doi.org/10.1016/j.marmicro.2012.11.001>
- Radomirović, M., Miletić, A., & Onjia, A. (2023). Accumulation of heavy metal(loid)s and polycyclic aromatic hydrocarbons in the sediment of the Prahovo Port (Danube) and associated risks. *Environmental Monitoring and Assessment*, 195(2), 323. <https://doi.org/10.1007/s10661-023-10926-2>
- Reimer, P. J., Austin, W. E. N., Bard, E., Bayliss, A., Blackwell, P. G., Ramsey, C. B., Butzin, M., Cheng, H., Edwards, R. L., Friedrich, M., Grootes, P. M., Guilderson, T. P., Hajdas, I., Heaton, T. J., Hogg, A. G., Hughen, K. A., Kromer, B., Manning, S. W., Muscheler, R., ... Talamo, S. (2020). The IntCal20 Northern Hemisphere radiocarbon age calibration curve (0–55 cal kBP). *Radiocarbon*, 62(4), 725–757. <https://doi.org/10.1017/RDC.2020.41>
- Rezaei, M., Mehdinia, A., Saleh, A., Modabberi, S., & Mansouri Daneshvar, M. R. (2021). Environmental assessment of heavy metal concentration and pollution in the Persian Gulf. *Modeling Earth Systems and Environment*, 7(2), 983–1003. <https://doi.org/10.1007/s40808-020-00913-8>
- Rochon, A., de Vernal, A., Turon, J.-L., Matthießen, J., & Head, M. J. (1999). Distribution of recent dinoflagellate cysts in surface sediments from the North Atlantic Ocean and adjacent seas in relation to sea-surface parameters. *American Association of Stratigraphic Palynologists Contribution Series*, 35, 1–146.
- Röder, K., Hantzsch, F. M., Gebühr, C., Miene, C., Helbig, T., Krock, B., Hoppenrath, M., Luckas, B., & Gerdts, G. (2012). Effects of salinity, temperature and nutrients on growth, cellular characteristics and yessotoxin production of *Protoceratium reticulatum*. *Harmful Algae*, 15, 59–70. <https://doi.org/10.1016/j.hal.2011.11.006>
- Sangiorgi, F., & Donders, T. H. (2004). Reconstructing 150 years of eutrophication in the north-western Adriatic Sea (Italy) using dinoflagellate cysts, pollen and spores. *Estuarine, Coastal and Shelf Science*, 60(1), 69–79. <https://doi.org/10.1016/j.ecss.2003.12.001>
- Satake, M., MacKenzie, L., & Yasumoto, T. (1997). Identification of *Protoceratium reticulatum* as the biogenetic origin of yessotoxin. *Natural Toxins*, 5(4), 164–167. <https://doi.org/10.1002/19970504NT7>
- Schlichting, E., & Blume, H.-P. (1966). *Bodenkundliches Praktikum* (p. 295). Paul Parey.
- Shahabi-Ghahfarokhi, S., Josefsson, S., Apler, A., Kalbitz, K., Åström, M., & Ketzer, M. (2021). Baltic Sea sediments record anthropogenic loads of Cd, Pb, and Zn. *Environmental Science and Pollution Research*, 28(5), 6162–6175. <https://doi.org/10.1007/s11356-020-10735-x>
- Siani, G., Paterne, M., Arnold, M., Bard, E., Métivier, B., Tisnerat, N., & Bassinot, F. (2000). Radiocarbon Reservoir Ages in the Mediterranean Sea and Black Sea. *Radiocarbon*, 42(2), 271–280. <https://doi.org/10.1017/S0033822200059075>
- Sönmez, İ, Kaplan, M., & Sönmez, S. (2014). Effect of chemical fertilizers on environmental pollution and its prevention methods. *Derim*, 25(2), 24–34.
- TÜİK. (2022). The results of address based population registration system, 2022. <https://data.tuik.gov.tr/Bulten/Index?p=Adrese-Dayali-Nufus-Kayit-Sistemi-Sonuc-lari-2022-49685>
- Turkoglu, M., & Oner, C. (2010). Short time variations of winter phytoplankton, nutrient and chlorophyll a of Kepez harbor in the Dardanelles (Çanakkale Strait, Turkey).

- Turkish Journal of Fisheries and Aquatic Sciences*, 10(4), Article Article 4.
- Ustaoglu, F., & Islam, Md. S. (2020). Potential toxic elements in sediment of some rivers at Giresun, Northeast Turkey: A preliminary assessment for ecotoxicological status and health risk. *Ecological Indicators*, 113, Article 106237. <https://doi.org/10.1016/j.ecolind.2020.106237>
- Ustaoglu, F., Tepe, Y., & Aydin, H. (2020). Heavy metals in sediments of two nearby streams from Southeastern Black Sea coast: Contamination and ecological risk assessment. *Environmental Forensics*, 21(2), 145–156. <https://doi.org/10.1080/15275922.2020.1728433>
- Ustaoglu, F. (2020). Ecotoxicological risk assessment and source identification of heavy metals in the surface sediments of Cömlekci stream, Giresun, Turkey. *Environmental Forensics*, 0(0), 1–13. <https://doi.org/10.1080/15275922.2020.1806148>
- Uzar, S., Aydin, H., & Minareci, E. (2010). Dinoflagellate cyst assemblages in the surface sediments from Izmir bay, Aegean sea, Eastern Mediterranean. *Scientific Research and Essays*, 5(3), 285–295.
- Velikova, V., Moncheva, S., & Petrova, D. (1999). Phytoplankton dynamics and red tides (1987–1997) in the Bulgarian Black Sea. *Water Science and Technology*, 39(8), 27–36. [https://doi.org/10.1016/S0273-1223\(99\)00184-5](https://doi.org/10.1016/S0273-1223(99)00184-5)
- Versteegh, G. J. M. (1994). Recognition of cyclic and non-cyclic environmental changes in the Mediterranean Pliocene: A palynological approach. *Marine Micropaleontology*, 23(2), 147–183. [https://doi.org/10.1016/0377-8398\(94\)90005-1](https://doi.org/10.1016/0377-8398(94)90005-1)
- Wall, D., & Dale, B. (1968). Modern dinoflagellate cysts and evolution of the Peridinales. *Micropaleontology*, 14(3), 265–304. <https://doi.org/10.2307/1484690>
- Wang, Y., Ling, M., Liu, R., Yu, P., Tang, A., Luo, X., & Ma, Q. (2017). Distribution and source identification of trace metals in the sediment of Yellow River Estuary and the adjacent Laizhou Bay. *Physics and Chemistry of the Earth, Parts a/b/c, Biogeochemical Processes in the Changing Wetland Environment*, 97, 62–70. <https://doi.org/10.1016/j.pce.2017.02.002>
- Wang, Z., Lin, K., & Liu, X. (2022). Distribution and pollution risk assessment of heavy metals in the surface sediment of the intertidal zones of the Yellow River Estuary. *China. Marine Pollution Bulletin*, 174, Article 113286. <https://doi.org/10.1016/j.marpolbul.2021.113286>
- Wignall, P. B., & Twitchett, R. J. (1996). Oceanic anoxia and the end Permian mass extinction. *Science*, 272(5265), 1155–1158. <https://doi.org/10.1126/science.272.5265.1155>
- Xie, R., Sun, D., Sun, L., Zhang, N., Zhang, J., & Zang, S. (2021). Sediment record of heavy metals over the last 150 years in Northeast China: Implications for regional anthropogenic activities. *Ecotoxicology*, 30(7), 1354–1365. <https://doi.org/10.1007/s10646-021-02350-4>
- Yang, W., Cao, Z., Zhang, H., & Lang, Y. (2021). A national wide evaluation of heavy metals pollution in surface sediments from different marginal seas along China Mainland. *Regional Studies in Marine Science*, 42, Article 101637. <https://doi.org/10.1016/j.rsma.2021.101637>
- Zhai, B., Zhang, X., Wang, L., Zhang, Z., Zou, L., Sun, Z., & Jiang, Y. (2021). Concentration distribution and assessment of heavy metals in surface sediments in the Zhoushan Islands coastal sea. *East China Sea. Marine Pollution Bulletin*, 164, Article 112096. <https://doi.org/10.1016/j.marpolbul.2021.112096>
- Zhang, L., Ye, X., Feng, H., Jing, Y., Ouyang, T., Yu, X., Liang, R., Gao, C., & Chen, W. (2007). Heavy metal contamination in western Xiamen Bay sediments and its vicinity. *China. Marine Pollution Bulletin*, 54(7), 974–982. <https://doi.org/10.1016/j.marpolbul.2007.02.010>
- Zhang, G., Bai, J., Zhao, Q., Lu, Q., Jia, J., & Wen, X. (2016). Heavy metals in wetland soils along a wetland-forming chronosequence in the Yellow River Delta of China: Levels, sources and toxic risks. *Ecological Indicators*, 69, 331–339. <https://doi.org/10.1016/j.ecolind.2016.04.042>
- Zhang, Y., Yu, J., Su, Y., Du, Y., & Liu, Z. (2019). Long-term changes of water quality in aquaculture-dominated lakes as revealed by sediment geochemical records in Lake Taibai (Eastern China). *Chemosphere*, 235, 297–307. <https://doi.org/10.1016/j.chemosphere.2019.06.179>
- Zhang, M., Sun, X., & Xu, J. (2020). Heavy metal pollution in the East China Sea: A review. *Marine Pollution Bulletin*, 159, Article 111473. <https://doi.org/10.1016/j.marpolbul.2020.111473>
- Zonneveld, K. A. F., Marret, F., Versteegh, G. J. M., Bogus, K., Bonnet, S., Bouimtarhan, I., Crouch, E., de Vernal, A., Elshanawany, R., Edwards, L., Esper, O., Forke, S., Grøsfjeld, K., Henry, M., Holzwarth, U., Kieft, J.-F., Kim, S.-Y., Ladouceur, S., Ledu, D., ... Young, M. (2013). Atlas of modern dinoflagellate cyst distribution based on 2405 data points. *Review of Palaeobotany and Palynology*, 191, 1–197. <https://doi.org/10.1016/j.revpa.2012.08.003>

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